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Master Thesis

Financial Risk-Factor Modeling: A comparison of Characteristic- and Statistical risk-factor models for forecasting portfolio risk in Norway

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Abstract

This master's thesis investigates the differences in performance generated from two different financial factor models in estimating risk in the Norwegian stock market. The research identifies a gap in transparency of estimation modeling among trading companies, which is why we developed a model for estimating risk in the Norwegian stock market. We developed a characteristic-based risk-factor model with factors expected to have significant explanatory power on the Norwegian stock market, and a statistical risk-factor model with abstract interpretation. We gather daily data from 2019 to 2025 historical price data and various financial report data. To compare these models, our aim was to bring models on equal footing with deliberate dynamic and static choices for both models. For evaluating both models ability to forecast portfolio risk, we generated global minimum variance portfolios of out-of-sample forecasts for both the characteristic and statistical factor model, as well as an addition naive sample covariance to serve as benchmark. We compare their OOS realized variance in a Diebold-Mariano test. We find that none of the models outperforms the other with statistical significance. However, we exclaim that study has many limitations such as measurement error and that the exponentially weighted covariance of the characteristic and statistical model does not in theory fulfill the assumption covariance stationarity made by the Diebold-Mariano test.

Preface

A long period of studies is now behind us, one of us part-time having two kids and working full time during the masters degree. The other full time studying.

Håkon would first like to thank his significant other for taking care of the kids by herself while i was writing in another room. I would also like to thank my employer for letting me study alongside working without having to worry about vacation.

Nicolay would like to thank his significant other and family for their unwavering support during the completion of the master's degree.

We also thank our supervisor, Christian Ewald, for his guidance during the completion of this thesis.

We are proud to present this thesis as the final result of our master's degrees.

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1 Introduction

1.1 Academic background to the study

Financial markets are increasingly and keenly concerned with the different aspects of risk among financial instruments throughout history. The financial markets are many, and its participants are also of great number, possessing varying levels of skill and tools in managing their own and other finances. Consumers, banks and investors all seek to lie ahead and have the best plan. However, future outcomes carry uncertainty, and the participants have become increasingly obsessed with information that allows for decision-making under uncertainty. All rational actors seek to minimize losses, and risk quantifies the chance of the realization of loss. Risk is a universal measure for all beings latently or consciously assessed in situations where a desired or undesired outcome is uncertain.

Most published articles within factor research are within the field of asset pricing. In asset pricing, factor models rely on e.g. agent-based, no-arbitrage, stochastic discount factor frameworks, and assumptions to establish inference. Rosenberg decomposes the CAPM beta to a linear combination of characteristic exposures and proposes an ex-ante market beta over Sharpe's ex post market beta (Rosenberg, 1974; Sharpe, 1964). Rosenberg (1974) makes no asset pricing related assumptions but rather proposes a methodology on asset risk decomposition. The two disciplines can therefore be boiled down to be defined as separate problems, namely asset pricing and risk management, respectively. Rosenberg's approach to risk decomposition inspired industry practitioners to adopt his modeling approaches to commercial use.

We decompose the methodology and research within fundamental equity risk models into several dimensions; formulation and postulation, data processing, and estimation, as well as evaluation. An essential example can be found in Connor et al. (2010) who presents Rosenberg's approach to factor modeling of security returns. Presenting a mathematical formulation of the estimation problem, Connor et al. (2010) also presents a comprehensive taxonomy of the asset pricing literature with observations and assessments of commonly included features / characteristics used in the generation of factor returns for equity risk models. Connor et al. (2010) presents the essentials of feature engineering such as standardization of characteristics, with market-capitalization weighted means so that exposures are compared in view of the total market (Connor et al., 2010). Paleologo (2025) provides a comprehensive introduction of both characteristic and statistical factors models, with extensive details on the estimation process and feasible solutions to challenges commonly encountered in fundamental risk equity risk modeling.

The main idea of behind characteristic factor models and their ability to explain co-variation in asset returns seemingly originates from Rosenberg and Marathe (1976), who

provided the core ideas in the characteristic-based factor representation of returns. While the original working paper is not widely accessible, but often cited as foundational, the core idea goes to show that firms can exhibit covariance or correlating structures reflected within a firm’s fundamental characteristics across cross-sections. Most of these ideas are also reflected in Rosenberg and McKibben (1973), and Rosenberg (1974). The intuition is that firms with similar microeconomic characteristics should be affected similarly by macroeconomic exposures. The standardized fundamentals in the cross-section can then be iteratively estimated across time to admit a low-dimensional factor covariance structure. A low-dimensional covariance structure has the benefit of being less prone to estimation error when compared to the often encountered high-dimensional covariance structure from pure asset returns. Risk-factor modeling has become a staple of commonly utilized methodology tools in the 20th century in modeling, analyzing and forecasting risk in the returns of financial instruments among practitioners and academics.

1.2 Explanation of key terms

Risk-factor model: A model that decomposes asset returns into exposures to common factors and an idiosyncratic component, allowing the estimation and forecasting of portfolio risk.

Characteristic-based (fundamental) factor model: A model where factors are constructed from firm-specific fundamental characteristics, such as size, value, or momentum, to explain the covariance structure of returns.

Statistical factor model (PCA): A model that extracts latent factors directly from historical return data without relying on economic or firm-level inputs, often viewed as a “black-box” approach.

Risk forecasting: The process of estimating the future variance and covariance of asset returns to support portfolio risk management.

Out-of-sample forecasting: Evaluating model performance using data that was not included in the model estimation, providing a test of predictive accuracy.

Covariance structure: The pattern of co-movement between asset returns, which factor models seek to estimate in a lower-dimensional representation.

Turnover: the change in portfolio weights from one period to the next.

We define “OSE all shares” as all equities that are available from our supplier of data and are traded on the Oslo Stock Exchange (OSE). These shares are traded on the main market called Oslo Børs (OSEAX), Euronext Expand Oslo and Euronext Growth Oslo.

1.3 Problem statement and research questions

In contrast to characteristic-based risk models, statistical approaches such as Principal Component Analysis (PCA) provide an accessible and nuanced alternative, discovering latent risk factors directly from asset return data. These models require no fundamental firm-level information and can therefore be implemented using only asset returns data, but at the cost of lacking economic interpretability. As a result, practitioners face a potential trade-off between economically interpretable factor structures collected from firm characteristics and purely statistical models that value computational simplicity and nuanced abstraction of driving forces in asset returns.

1.3.1 Main Research Question

Does a characteristic-based risk-factor model yield more accurate out-of-sample (OOS) covariance forecasts compared to a statistically estimated risk-factor model (PCA) within the Norwegian Equity market?

1.3.2 Hypotheses and Sub-Questions

To address the main question, our comparative framework is in the spirit of Engle and Colacito (2006), where we construct a *global minimum variance* (GMV) portfolio from each respective covariance matrix from the two models. In addition, we include a "naive" sample covariance matrix in our comparison to evaluate model specification. We compare the OOS-realized variance observed for each day generated from each portfolio, and compare them through a Diebold-Mariano hypothesis test.

1.4 Positioning

This thesis is positioned within the broader literature on factor models in finance but focuses specifically on their approach on risk estimation and covariance forecasting rather than asset pricing. While factor models have been widely studied in financial economics, much of the academic literature has primarily examined their role in explaining expected returns and testing asset pricing theories.

In the asset pricing literature, factor models are commonly developed within conceptual models like the stochastic discount factor approach, no-arbitrage models, or equilibrium-based asset pricing models. These frameworks aim to identify systematic sources of risk that explain cross-sectional differences in expected asset returns. Consequently, empirical factor models are typically assessed based on their ability to explain average returns rather than their effectiveness in forecasting future risk.

Unlike other approaches, risk modeling research looks at breaking down the covariance

structure of asset returns to estimate and predict portfolio risk. Rosenberg (1974) and Rosenberg and Marathe (1976) were among the first to use characteristic-based methods for modeling security returns, using firm-level characteristics to find common risk factors. Instead of depending on asset pricing assumptions, these models try to create a simple way to represent return covariance for managing portfolio risk.

Both areas use factor representations of asset returns, but they have different goals. Asset pricing models aim to explain expected returns and find sources of risk that affect prices. Risk models, on the other hand, focus on estimating and predicting how asset returns move together. In practice, especially in portfolio management, factor models are often used as tools for estimating risk instead of testing economic theories.

This thesis focuses on how well factor models perform in forecasting risk. It compares a characteristic-based factor model with a statistical factor model using principal component analysis to determine which yields better out-of-sample covariance forecasts. By focusing on risk forecasting instead of asset pricing, this study contributes to research on how factor-based equity risk models are used in practice.

1.5 Contribution – practical and theoretical implications

Fundamental risk models that use characteristics are the industry standard for companies such as MSCI and Barra. However, the details of their frameworks, especially how they handle data and build factors, are kept private. Their estimation methods are also not usually shared. Because of this lack of transparency, it is difficult for others to evaluate or replicate these models, whether in academic research or for professionals who do not use commercial risk management tools.

The lack of publicly accessible implementations creates a gap between the theoretical research on factor modeling in academia and the practical risk modeling structures used in industry. Academic literature has extensively studied factor models within the context of asset pricing; fewer studies present transparent and operational implementations aimed specifically at forecasting risk and covariance frameworks in real markets. As a result, practitioners seeking to construct or evaluate risk models using publicly available data face considerable uncertainty regarding the effectiveness of alternative methodologies. We exclaim the difference between asset pricing and risk forecasting as these are separate objects of inference, we focus primarily on risk forecasting but note that much of its foundations are derived from seminal contributions in the field of asset pricing.

This thesis contributes to the literature by developing and evaluating a transparent implementation of a characteristic-based equity risk model using publicly available financial and market data. By constructing such a model and comparing its forecasting performance to that of a statistical factor model estimated through Principal Component Analysis

(PCA), the study provides empirical insight into the trade-off between economically interpretable factor structures and purely statistical approaches to covariance estimation. The comparison allows for an assessment of whether the additional economic structure embedded in characteristic-based models leads to improved out-of-sample risk forecasts.

As well as methodological contribution, the thesis presents empirical evidence from the Norwegian equity market, a relatively small and less liquid market compared to the large international markets where many factor modeling frameworks have traditionally been developed and tested. The performance of different risk modeling approaches in such markets remains less documented in the literature. By evaluating the predictive performance of both models in this context, the study adds to a better understanding of how risk modeling techniques perform in smaller and potentially less efficient markets.

Taken together, the findings of this thesis may be of interest both to academic researchers studying factor models and to practitioners seeking transparent and implementable approaches to equity risk modeling.

1.6 Choice of a research context

This thesis sits within the larger body of work factor models in finance, but it specifically looks at how they handle risk estimation and covariance forecasting instead of asset pricing. Factor models have received a lot of attention in financial economics, but much of the research mainly focuses on their ability to explain expected returns and test asset pricing theories.

In smaller and less liquid markets like Norway's, it is still uncertain if the extra complexity and data needed for characteristic-based factor models actually improve risk forecasts compared to statistical methods. Characteristic-based models use detailed company information and structured factors, while statistical models such as principal component analysis find risk factors directly from return data without needing fundamental inputs.

The Norwegian market therefore provides a relevant context for evaluating whether the added economic structure embedded in characteristic-based models translates into improved out-of-sample covariance forecasts, or whether simpler statistical models are sufficient for capturing the underlying risk structure of asset returns in smaller markets.

1.7 Overview of how the thesis is structured

The theoretical framework presents an overview of theory that discusses the implications of prediction assuming that within asset returns there exists a linear factor structure that drives returns systematically and unsystematically. The methodology presents the linear factor structure and how it relates to portfolio risk, in addition to methods of estimation

for both the characteristic and statistical model. We formalize the inner workings of hypothesis tests that will compare both models. In the methodology chapter, we will also provide an extensive data description documenting how the data is processed, where significant corrections were made, in addition to a discussion regarding the integrity of the data and the data vendor.

2 Theoretical Framework

2.1 Foundations and seminal papers in Factor Modeling

We present the transition from the linear factor structure pioneered by the field of asset pricing to operationalized risk-factor models which map and survey various sources of portfolio risk.

The theory emerges from the foundational findings of Ross (1976). Ross assumes that financial markets preclude arbitrage and proposes that asset returns follow an approximately linear factor structure. Financial asset returns are admitted and presented as linear functions of factor risk exposures, in addition to an asset specific or idiosyncratic component with zero-expected returns and finite variance. Note the decomposition of these two main sources asset risk. Systematic risk is undiversifiable and can be thought of as an embedded covariance structure emerging from market business cycles or macroeconomic determinants. Practitioners also characterize a few systematic risk sources as anomalies, due to limited economic or rational interpretation. Idiosyncratic risk is asset specific, and pays no risk premium as it is free for the investor to diversify away by including more assets, hence establishing a portfolio. Ross (1976) generalizes seminal results by Sharpe (1964), and imposes fewer restrictions on the statistical moments of returns than the *Capital-Asset-Pricing-Model* (CAPM) equilibrium model.

Chamberlain and Rothschild (1983) extends Ross (1976) no-arbitrage arguments, where one of the key results is that PCA has theoretical grounds in uncovering systematic risk structure. They propose that as more assets are added to the investment universe, specific eigenvalues approximately spike and others stay bounded, they show that the bounded eigenvalues can be attributed to idiosyncratic risk. Chamberlain and Rothschild (1983) imposes weaker constraints on the linear factor structure allowing the idiosyncratic components of portfolios to exhibit correlation. This is in contrast to Ross (1976) who imposes the strict condition that the idiosyncratic component of portfolios be uncorrelated. Rosenberg (1974) addresses the issue of CAPM yielding an estimate of the historical beta, Rosenberg proposes that the market beta can be estimate ex ante as a linear combination of asset descriptors, *e.g.* accounting data or other characteristics, the proposition follows that descriptors of assets encode covariance structure of economic determinants among assets.

It was soon discovered that firm characteristics were sufficiently practical to act as coefficients or exposures to these determinants or factors. Grinold and Kahn (2007) connects the aforementioned models, the characteristic model of Rosenberg ¹ and the statistical

¹who founded the financial consulting company BARRA where one of the main services was the design and production of financial risk models Rosenberg (2026).

model to Ross (1976) *Arbitrage-pricing-theory* (APT). Grinold and Kahn (2007) states that these models qualify as APT models. This is due to their linear factor structure and the connection they share with the findings of Ross (1976).

Why forecast portfolio risk assuming a linear factor structure? Portfolio risk can be decomposed to:

$$\Sigma_{\mathbf{R}} = \mathbf{B}\Sigma_{\mathbf{F}}\mathbf{B}^{\top} + \Sigma_{\mathbf{E}}$$

Recall, that in well-diversified portfolios idiosyncratic risk dissipates as more assets are added to the portfolio. The low-rank factor structure dominates in signal over the idiosyncratic covariance matrix, and the constructed asset covariance matrix becomes more well behaved in contrast to estimating the empirical sample covariance matrix of asset returns. In machine learning, practitioners would recognize this as a trade-off between bias and variance. Where the linear factor structure yields some bias, but reduces variance in estimation significantly.

Using characteristic-based factors does not always provide useful information. John H. Cochrane (2011) points out that the growing number of proposed factors in the literature may harm the feasibility and credibility of findings. Researchers need to be careful when selecting characteristics, as not all documented factors reflect true sources of systematic risk.

2.2 A Taxonomy of Known Risk Factors

Table 1. Taxonomy of Financial Factors in Risk-Factor Modeling

Factor	Proxy	References
Value (Book-to-Market)	$\frac{\text{Book Equity}_{i,t}}{\text{Market Cap}_{i,t}}$	Daniel and Titman (1997) and Fama and French (1993)
Size	$\ln(\text{Market Cap}_{i,t})$	Fama and French (1993) and Van Dijk (2011)
Momentum	$\prod_{d=t-252}^{t-21} (1 + r_{i,d}) - 1$	Carhart (1997) and Jegadeesh and Titman (1993)
Weekly Reversal	$-\sum_{d=t-5}^{t-1} r_{i,d}$	Lehmann (1990)
Log of Idio. Vol.	$\ln(\text{Var}(\hat{\varepsilon}_{i,t}^m))$	Ang et al. (2006)
Rolling Illiquidity	$\frac{1}{K} \sum_{d=t-K+1}^t \frac{ R_{i,d} }{\text{NOK Volume}_{i,d}}$	Amihud (2002)
Dividend Yield	$\frac{\sum_{d \in [t-252, t]} \text{Dividends}_{i,d}}{\text{Price}_{i,t}}$	Naranjo et al. (1998) and Zimmer (2024)
Oil Beta	$\hat{\beta}_i^{\text{Oil}} = \frac{\text{Cov}(r_{i,t}, r_{\text{Oil},t})}{\text{Var}(r_{\text{Oil},t})}$	Chen et al. (1986)
Market Beta	$\hat{\beta}_i^m = \frac{\text{Cov}(r_{i,t}, r_{m,t})}{\text{Var}(r_{m,t})}$	Frazzini and Pedersen (2014) and Sharpe (1964)
Rate Sensitivity	$\hat{\beta}_i^y = \frac{\text{Cov}(r_{i,t}, r_{y,t})}{\text{Var}(r_{y,t})}$	Chen et al. (1986)
Growth	$\frac{\Delta \text{Total Assets}_{i,t}}{\text{Total Assets}_{i,t-1}}$	Cooper et al. (2008)

2.2.1 Factor explanation

The purpose of including these factors is not to test whether they earn premia, but to determine whether they improve the modeling of covariance and total risk in Norwegian equities. Each factor is therefore included because it captures a distinct source of systematic variation in returns. Rather than viewing these variables purely as return predictors, we interpret them as proxies for underlying economic risks or structural characteristics that predict the return direction of stock prices. Some of the characteristics are documented anomalies across global equity markets. A significant portion of the samples in asset pricing research is however dominated by North American equities, and whether all of them fit the Norwegian Equity scene is a matter of question for many of the characteristics. We assume that some of these anomalies are transferrable characteristics from North American equity markets, can aid in explaining the cross-section of returns in Norwegian stocks. We will provide both arguments and reference documentation in the following subsections. We also argue that other, less documented characteristics fit the

Norwegian equity market narrative more than other foreign markets, such as oil sensitivity and illiquidity. There is reason to believe that dividend yields fit Norwegian equity markets more than others, due to Norway's shielding tax on dividend returns generated (Zimmer, 2024).

2.2.2 Value (Book-to-Market)

The value factor captures the tendency of stocks that are considered an undervalued stock or an overvalued stock. These categories of stock tend to rise together or fall together. Including this factor allows the model to capture that covariance component. Value stocks are firms with low market value compared to accounting value. If this reflects risk compensation or systematic mispricing is still debated in the literature.

To find the value, we divide the company's total book equity per share by the share price. We must also distinguish between our use of Book-to-Market to generate factor returns, and that of Fama and French (1993) value factor, which uses the same Book-to-Market measure, but in contrast, uses portfolio sorts to generate factor returns. It is also important to note that Fama and French (1993) drops negative Book-to-Market assets in their portfolio sorts. We retain negative Book-to-Market, as these tend to cluster during regimes of economic distress, carrying information of covariation among equities.

2.2.3 Size

Banz (1981) documents the size effect. The anomaly shows that smaller firms earn a higher risk-adjusted return, than larger firms. In a working paper by Næs et al. (2010), they find that the anomaly persists in the Norwegian equity market from 1980 to 1999. Small-cap stocks also tend to have more financial pressure in the shape of debt, which in turn makes small-cap stocks more sensitive to economic drops. In risk-modeling, the size factor takes two important aspects into account; Liquidity affects how hard it is to sell the stock if there is a market crash. Exposure to funding limitations, which will be a decreasing problem, the larger the firm becomes. The size factor is primarily used to capture differences in firm resilience.

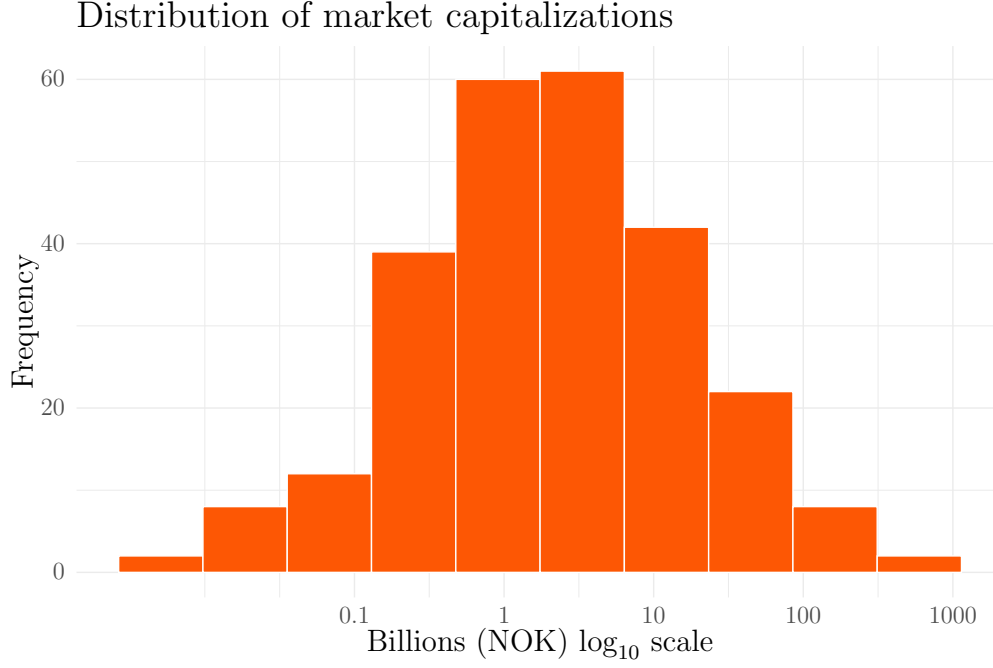


Figure 1. The sample distribution of market capitalizations for Oslo stock exchange at 30-12-2025 in $\log_{10}$ scale. The scaling helps in visualizing the highly skewed distribution of market caps.

We observe in Figure 1. that the our sample of equities from Oslo stock exchange sum of market capitalizations is dominated by large companies which Næs et al. (2010) also notes. Companies such as Equinor, Aker BP, DNB and Hydro, to name few, makes the distribution of market capitalizations on the Oslo stock exchange highly skewed, which can be viewed in the figure below, noted in $\log_{10}$ scale ².

2.2.4 Momentum

Momentum captures the tendency of stocks with strong past returns to continue performing well in the short-to-medium term. This effect was documented by Jegadeesh and Titman (1993) and incorporated into factor models by Carhart (1997).

Momentum may reflect delayed information diffusion, behavioral underreaction, or compensation for crash risk. From a risk perspective, momentum portfolios exhibit distinctive covariance structures, especially during market reversals.

2.2.5 Short-term Reversal

Short-term reversal captures mean reversion at very short horizons, as documented by Lehmann (1990). This factor is often attributed to liquidity provision, bid-ask bounce,

²Without the scaling, we would observe Equinor as a very small observation located to the very far right significantly separated from other firms.

and temporary price pressure. It is particularly relevant in risk models to account for microstructure-driven return predictability.

2.2.6 Idiosyncratic Volatility

Idiosyncratic volatility measures firm-specific risk unexplained by systematic factors. Following Ang et al. (2006), idiosyncratic volatility is computed as the standard deviation of CAPM residuals.

Although idiosyncratic risk should be diversifiable in theory, empirical evidence suggests it may be priced or associated with limits to arbitrage. In a risk-modeling framework, it is central to decomposing total volatility into systematic and specific components.

2.2.7 Illiquidity

Liquidity is included to capture systematic variation in trading conditions and funding constraints. In smaller equity markets such as Norway, liquidity shocks may generate substantial co-movement across firms, particularly during stress periods. Including a liquidity proxy therefore improves the model's ability to capture state-dependent covariance dynamics.

Firms with similar liquidity characteristics tend to co-move during stress episodes, for example, illiquid shares tend to fall further than liquid shares due to the fact that they are harder to offload.

2.2.8 Dividend Yield

Dividend captures firm maturity and payout policy. When a firm has matured, it usually produces money that does not go into growth, but to their investors. Growing companies tend to place their profits into expansion instead of dividends. In recessions, dividend paying firms tend to be more defensive than growth firms, making fewer changes to their operations because they bring back profits. High-yield stocks often "move together" because they attract the same demographic of investors.

Norway allows retail-shareholders to receive a portion of their dividends tax-free. This shielding allowance amount is calculated based on a risk-free interest rate applied to the cost basis of the shares. The shielding allowance makes dividend yield even more important to add to the model because if high yield stocks move together in countries where these profits for investors are taxed, then we believe that this covariance will be even clearer on Norwegian stocks.

Using trailing dividend yields, it is possible to observe equities which display dividend yields above 1, this is not necessarily unrealistic, as stock prices can collapse during the

trailing 12-month period. A caveat with dividend yields is that firms with yields above largely above are possible, and most often reflect distress due to price collapse, this makes the regressor non-linear.

2.2.9 Oil Beta

Given the importance of the petroleum sector in Norway, exposure to oil price movements represents a systematic macroeconomic risk factor. Oil price shocks affect Norwegian economic activity, exchange rates, and corporate earnings. Capturing oil sensitivity improves risk decomposition in the Norwegian market. Firms with high oil sensitivity share exposure to a common macroeconomic risk. Næs et al. (2010) claims that there is limited empirical evidence on oil being a priced risk-factor in Norway. However, OSE's significant concentration of petroleum industries may provide through rationale for using this as a feature. The oil beta could proxy international risk as well.

2.2.10 Market beta

Based on the CAPM of Sharpe (1964), market beta measures sensitivity to aggregate market movements. Even in multi-factor models, market beta captures broad macroeconomic shocks that affect all firms. It serves as the baseline systematic risk component.

2.2.11 Yield sensitivity

Interest rate sensitivity captures duration effects and exposure to monetary policy shocks, particularly relevant for financials and capital-intensive industries. Δy_t represents the changes in government bond yields. This factor is also important for capturing changes in interest rates compared to dividend yields. This way, we can determine if dividend yields change because of a change in interest rate or if there are other reasons. Chen et al. (1986) shows that the yield spread or curve dynamics has significant explanatory power on asset returns. We chose not to model the spread, but to rather use the basis point changes in the 10 year Norwegian government bond yield. The main reason is that it serves as a discounting factor for firms, as well as incorporating the interest rate expectations in the Norwegian economy.

2.2.12 Growth

In a risk model, investment captures differences in corporate policy and growth intensity that generate systematic co-movement. This factor is connected to dividend yields as well due to the fact that the firms that pay out dividends do not spend it on expanding.

2.2.13 Industry/sector not included in the model

Grinold and Kahn (2007) demonstrates that the majority of factors in BARRA models are industry indices. We did not have access to proper industry data, but we did have access to the sector affiliation of firms. Which is unfortunate, since we believe adding sector affiliation would make the model more robust. The reason it was not added, was because later in the master thesis timeline (before OOS evaluation was executed) some assets reflected sector affiliation/transition not known at the time. We decided the potential for data leakage was greater, which is why we decided to drop the binary table of sector affiliations containing 13 sectors.

3 Methodology

3.1 Choice of research design

3.1.1 Models

PCA is a widely used quantitative risk modeling tool, especially for estimating covariances from return data. However, it raises the question of whether risk estimation can be improved by using additional information about the assets or whether interpretability is worth the sheer amount of data. Fundamental factor models can create more intuitive factors than PCA.

Estimates like these are primarily used for portfolio construction. Comparing these two models is therefore executed in a portfolio setting, using the same investment universe in both models. In addition, fundamental factors may provide a more intuitive understanding of how different sources of risk affect the portfolio.

Because the models are intended for portfolio use, covariance estimates are treated as forward-looking and applied to out-of-sample portfolio construction. This approach matches practical use, where investment decisions rely on expected future risk rather than solely on historical data.

3.2 Choice of data collection and analysis method

3.2.1 Dataset

We have chosen to research the assets traded on the Oslo stock exchange and have gathered data from 2nd of January 2019, to 30th of December 2025. The data includes daily close prices, dividend dates, dividend yields, and quarterly balance sheets on each asset denominated in NOK.

Why add Euronext growth and expand? The Norwegian equity market is small, consisting of a few firms with large market capitalization, and many firms with small market capitalization.

The main data was collected from EODHD. From EODHD, we have collected reported book equity, shares outstanding, total assets, total liabilities, and report publishing dates from balance and earnings sheet data. Report publishing dates were in the majority cases, often not present in the balance sheet data, but were however often present in earnings data. We have therefore made the assumption that the earnings reports of all equities were released simultaneously with balance sheet reports. We have also retrieved daily trading volume, historical close- and adjusted close prices, and a list of listed and delisted tickers metadata on Oslo Stock Exchange (Euronext all-shares index) for filtering and cleaning duplicates, mergers, symbol-changes and share-classes. For historical close prices, both adjusted and unadjusted we were able to retrieve a time-series of 532 tickers. This time-series contained an excessive amount of missing values, most often these values will prove to be structural, due to firms listing and delisted on Oslo Stock Exchange. The inclusion of both active and inactive assets is meant to counter the sample selection bias of survivorship. If we used active assets exclusively, our distribution of returns would be positively skewed. In very few cases, some shares had different share classes, and we opted for the use of A classes, which had more trading volume. The sub-classes of shares were therefore deemed redundant information and would likely contribute to generating correlated estimation error, which would lead to inefficient model estimation. These different classes of the same symbol's shares were discovered through excessive correlation analysis, with coefficients yielding high values, *e.g.* $\rho \approx 0.99$ between the corresponding share classes.

We have collected data from Yahoo Finance, which aims to aid in feature construction. From Yahoo Finance we have obtained Brent oil futures close prices and various currency pairs. EODHD has limited data available for currency pairs and Brent oil futures, which is why we have resolved to obtain the data from Yahoo Finance. The currency pairs have been used for adjusting report values denoted in foreign currencies to the common currency of NOK. Most values within the balance sheet data were labeled with the correct currency. However, in a few cases, some tickers had incorrect currency labels, which made

automated data processing challenging. Discovered through outlier detection, we label this subset of tickers: Broken data, but we do not disclose the possibility of broken data passing through personal inspection and outlier detection. However, the sample size is restricted to how well both models can handle missing data. For example, assuming that the idiosyncratic volatilities of firms are uncorrelated, we estimate the exponentially weighted (EW) idiosyncratic variance with the same half-life implied in the smoothing parameter of the EW-covariance of both the characteristic and statistical model. So we set two main rules for sampling: 1. we require that any firm is alive at the latest in-sample observation point. 2. In addition we require that any firm has at least a history of 135 days. The rule is somewhat arbitrary, 10×13.51 , where $\text{half-life}_\lambda = 13.51$, but it ensures sufficient history. One could argue that this is in favor of the characteristic model, since it requires estimates of idiosyncratic volatility for not underestimating risk. However, this also benefits the statistical model in the sense that it requires less imputation. Imputation of missing values is especially severe when missing values are structural and meaningful, *e.g.* listings and de-listings.

The betas

3.2.2 Model selection

The characteristic-based model is built on empirical asset pricing theory and creates factors from observable firm characteristics. This means the factors can be understood in economic terms, such as company size or valuation. Unlike statistical methods, researchers can choose the number and type of factors based on what mattered in past studies. This model helps include meaningful economic information when estimating covariances.

The statistical model uses PCA to find factors directly from the return data. These factors are not linked to economic variables and are only indirectly interpretable. PCA provides an efficient statistical summary but does not reveal much about the economic causes of risk.

The comparison of both models tests and compares the prediction capabilities of imposing economic and statistical structure on financial return data.

The factors in the two different models will differ in the sense that the characteristics will have predetermined factors, while the PCA will determine factors based on the return data gathered. The number of factors will vary on the PCA model because of the opportunity to cancel noise. While on the characteristic model, the weights of the factors will vary in each cross-section across the assets.

3.2.3 Rolling SVD and Cross-sectional WLS

PCA is estimated on a window of 252 trading days; in other words a trading year, and singular value decomposition (SVD) is applied. This prevents the use of future information broadly stationary. It also gives the factor loadings the opportunity to adapt to changes in market conditions.

3.2.4 Covariance estimation

We use exponential weighting scheme on both factor models to weight the most recent history of assets. Financial volatility has shown that different assets move together when conditions are more uncertain. This strengthens the importance of recent information and lightens the weight of older information, which may have been a completely different financial situation in the market. The objective is to forecast how the covariance moves, which surfaces the need for a model that adapts to the underlying data. The need to capture the latest market fluctuations is prioritized over the risk of the model becoming unresponsive if larger events were to affect the market in the short term. We call the smoothing parameter θ , we have reserved the more common notation λ for eigenvalues. We have set the smoothing parameter. The characteristic based factor-covariance is estimated through the R package *PerformanceAnalytics*, which has an EWMA function for estimating the exponentially smoothed covariance. For the characteristic idiosyncratic variance we used the *forecast* package's simple exponential moving average to estimate the idiosyncratic variances. The idiosyncratic variance is assumed to be uncorrelated. The naive sample covariance is calculated using a 252-day window with pairwise observations.

3.2.5 Forecasting

We use all the data available as the training set and start the out-of-sample as soon as the models have enough data. The first day forecast is February 21st, 2020. Behind this date is an estimation window of 252 days. We use EWMA to forecast factor covariance one day ahead using exponential weighting and a half-life of 13.5 days. The model will update continuously and use the day just forecast to enable one-step-ahead forecasts. In other words, the model uses information up to time t to forecast the covariance at time $t + 1$.

3.2.6 Bias vs. Variance

The covariance matrix is the primary object of estimation for forecasting and attributing risk. To achieve accurate risk forecasts, the problem statement can be reduced to a common goal in machine learning, namely to seek out estimates with the lowest prediction error according to a prespecified loss function. The prediction error consists of reducible

and irreducible error.³ In a prediction problem, the prediction error that is reducible by methodological approaches can be decomposed to consist of bias and variance (James et al., 2021). The *bias* can be a systematic error from an incorrectly specified model structure, *e.g.* it is often argued that CAPM is insufficient in explaining most variation in equities, and that more systematic sources of risk exist and must be accounted for. Therefore, when the single-factor assumption is often rejected, and the CAPM would suffer omitted variable bias. In our case, we are making similar assumptions, namely that the covariance structure in returns can be explained through 13 factors. *Variance* arises from the fact that model estimates vary between inputs, *e.g.* pronounced variance in the context of the covariance matrix will lead to substantial variation in estimates, hence the term variance. A practitioner is certain to encounter the variance issue when engaging in estimation from high-dimensional data. Bias and variance both have detrimental consequences in applied portfolio construction. In terms of empirical covariance estimation, the estimation quickly becomes complex if the number of assets and observed returns are approximately the same in number. The empirical covariance matrix is unbiased, but high in variance when applied to high dimensional data.

³Irreducible error can be defined as inherent noise in the data, which intuitively enough, cannot be reduced without adapting predictions to noise.

3.3 Variable/feature construction and model inputs

3.3.1 Characteristic factor model

For a single cross-section, the characteristic-based risk-factor model can be represented as the following,

- $R_t \in \mathbb{R}^N$: is a vector of excess returns from assets at time t .
- $B_t \in \mathbb{R}^{N \times K}$: is a matrix of firms N by standardized descriptors or exposures K at time t .
- $f_t \in \mathbb{R}^K$: is a vector of the returns of individual factors at time t .
- $\epsilon_t \in \mathbb{R}^N$: vector of idiosyncratic returns at time t .

The returns and risk of assets is defined and explained by the structure respectively:

$$R_t = B_t f_t + \epsilon_t$$

$$\Sigma_R = B \Sigma_F B^\top + \Sigma_E$$

The methodology presented by Rosenberg was to assume the characteristics of equities encode covariance structure. The shared descriptors among firms are therefore assumed to form latent factor returns, which also shares a low-dimensional covariance structure. In contrast to Fama & French (1993) who establishes factor returns on portfolios sorts and their historical returns. Rosenberg (1974) estimate factor returns statistically and treat the factor loadings or coefficients as known and reflected through firm descriptors. The object of the estimation in the previous representation is therefore not the factor sensitivities X_t , but the factor returns f_t . This implies several advantages over historically estimated Fama-French factor models. Exposures are interpretable, that is, if the value factor or a particular industry has greater exposure $X \gg 0$ in a portfolio, the portfolio manager has essentially established a long-position in high B/P firms. The exposures are time-varying, forward-looking, and updated as soon as new information arrives to the public.

Firms are heterogeneous on several occasions, size, volatility, earnings, etc. It is therefore implied that small firms are noisy and carry large idiosyncratic variance, in contrast to large firms which are aggregated and less noisy. *Ordinary-Least-Squares* (OLS) becomes inefficient in the presence of heteroskedasticity; *Generalized-Least-Squares* (GLS) becomes a more efficient estimator of factor returns. In GLS, the weighting matrix W would yield unbiased estimates if we knew the true residual covariance matrix. However, we do not know it advance, and a common approach is to proxy the true residual covariance matrix

by setting $W_m = \sqrt{(\text{Market cap.})_i}$ the weighting matrix equal to the squared market capitalization of firms. While a useful proxy, it does not guarantee efficiency in estimation. Grinold and Kahn (2007) suggests that this is an appropriate weighting scheme, Paleologo (2021) show that it is implicitly assuming that the idiosyncratic variances of firms are proportional to the inverse square root for the market capitalization of firms, he also suggests using averaging the market cap over a 21-day window. It is not mentioned why it is useful, but a professional guess would be that the 21-day equal weighted average window serves as filter for microstructure noise.

The factor returns are estimated through:

$$f = (B_t^\top W_t B_t)^{-1} B_t^\top W_t R_t$$

Where each regression is performed on a cross-section. Each regression returns a factor return for the corresponding cross-section. These same factor returns are in turn used to calculate the characteristic-based factor covariance matrix. The factor return covariance has also received a bias-correction based on the fact that estimated factor returns covariance is biased upward because its estimates are based on estimates. We derive the bias correction in the appendix in section B.

3.3.2 Statistical factor model

The statistical model is based on the techniques of principal components analysis. Which is a dimensionality reduction method which identifies directions or components that explain the maximum amount of variation in the data. The theory of the model relates to the APT latent factor structure presented by Chamberlain and Rothschild (1983), they present approximate factor structure which allows for correlated idiosyncratic returns, and strict factor structure which assumes residuals or idiosyncratic returns are uncorrelated across all assets. We have chosen a strict factor structure, allowing for the use of relatively simple methods and estimation procedures for model construction. We apply principal component analysis on a 252-day window, Paleologo (2025) notes that the width is chosen so that the model can be considered broadly stationary. For statistical model construction, Paleologo (2025) presents several schematics and modeling choices, to how a practitioner might go about creating a statistical factor model on their own. We found few R-packages that resemble the baseline of the statistical factor modeling process, while maintaining an "apples to apples" guided decision-making for comparing a characteristic factor model to the statistical factor model. We have therefore made our own. The construction process have been adapted and documented from Paleologo (2025), and is available in the appendix as both a conceptual guide and R-code.

There exists a variety of ways to select the number of factors for a statistical factor model. A key result from random matrix theory is that noisy covariance matrices produce eigenvalues or factors below the Marchenko-Pastur thresholds, in other words one can confidently claim that noise gathers within a specific interval among eigenvalues, which are these exact noise thresholds. When assumptions are met, it is possible to use this threshold to determine the number of factors that spike above the noise bulk. We found the assumptions of We therefore settled on model goodness-of-fit diagnostics, such as penalty-based methods using the information criteria suggested by Bai and Ng (2002), which suggests a number of factors with the aim of balancing model fit and complexity, in a similar fashion to widely known diagnostic measures such as Aikake- and Bayesian information criterions (IC). Bai and Ng (2002) finds that their presented IC2 identifies the true factor structure more correctly than their proposed IC1 and IC3. They find that IC3 often over-parameterizes and selects more factors than the true number present in their data. Overparameterization in our case may be a desired trait, as Paleologo (2025) suggests that selecting a few factors within the context of portfolio optimization underestimates true risk. Since this thesis aims to construct GMV portfolios, in order to uncover the best conditioned covariance matrix for volatility forecasting, we see the benefit of choosing more factors rather than less, and overestimate rather than underestimate. This led to the choice of IC3 as our main loss function for selecting the number of factors. This method is also flexible in the sense that it changes with the data inside the rolling window. This method also penalizes overfitting, excluding factors that seemingly provide noise.

We impute because the model requires a matrix with a value for every observation. Imputation yields the best results when the model has few missing values, because it uses the rest of the data to determine which values are likely to be present in the rest of the data. We use SVD-imputation to determine the imputation values. The imputation is used to fill missing data, which represent 0,04% of the data. Troyanskaya et al. (2001) finds that the SVD based imputation method perform best on time-series on genetic data. We utilize the `pcaMethods`-package in R to perform SVD-imputation. We provide more thorough details on construction in the appendix at section A.

3.3.3 Global Minimum Variance and Evaluation Criteria

We measure forecast performance by comparing and forming two individual GMV portfolios established on the basis of covariance forecasts generated by the characteristic factor model and the statistical factor model, respectively. The aim of the GMV portfolio is to produce the portfolio that achieves the lowest possible variance. In matrix algebra, the quadratic optimization problem can be defined as:

$$\min \mathbf{w}'\Sigma\mathbf{w}$$

subject to $\mathbf{1}'\mathbf{w} = 1$. Solving the problem yields the solution:

$$\mathbf{w} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}'\Sigma^{-1}\mathbf{1}}$$

We observe that solving for the GMV portfolio weights is dependent on the inverse of the covariance matrix, also known as the precision matrix. One may hypothesize that the performance of the GMV portfolio depends on the quality of the constructed covariance matrix. Engle and Colacito (2006) demonstrate that a correctly specified covariance matrix will yield the smallest volatility for a GMV allocated portfolio. Our paper will present a similar comparison method to that of Engle and Colacito (2006), we define our loss function as the squared return of the GMV portfolio, we compare the pairwise loss differentials between GMV portfolios formed by the characteristic and statistical factor model, in addition to the naive sample covariance benchmark utilizing a Diebold-Mariano test (Diebold, 2024; Diebold & Mariano, 1995). As mentioned the process of generating the portfolio weights are dependent on the inversion of the covariance matrix. Both the characteristic and the statistical model allows for Woodbury inversion (Paleologo, 2025), which is a structural advantage over the naive sample covariance.

3.4 The data collection and construction process

3.4.1 Data sources

We have used EODHD for the main dataset providing closing prices daily and balance sheets for every ticker on OSE all shares. A supplementary data provider has been Yahoo Finance which has Crude Oil future closing prices and currency pairs, in addition to the Norwegian Central Bank's historical yield on 10-year government bonds for construction of the yield beta, and the Norwegian weighted average Overnight Financing Rate used in the construction of a risk-free rate for excess return calculation for the various betas.

3.4.2 Sample selection and construction

OSE all shares constituent asset returns are the dependent variables of our independent regression estimation. The individual asset returns are derived from end-of-day close prices. Close prices are assumed to be reflective of the true price, given that end-of-day prices are subject to closing auctions which are liquid events that often account for some of the largest volume traded on a given day (REF).

The individual unadjusted asset close prices will be utilized to form accounting features and factor returns. We have chosen unadjusted close prices for feature generation as they

do incorporate information regarding future corporate events, such as splits or dividends.

The dataset is small, and we add those to make it as big as possible. We use the square root of market cap to generate factor returns and weight the market cap of each ticker after how high the market cap is. The higher the market cap, the more weight on the estimation of idiosyncratic volatility.

All equities registered on OSE are obligated to report quarterly. However, note that the LOCF imputation method does not forward observations any longer than a quarter. The latter is a deliberate choice to avoid exacerbating measurement error. How we solve reports that appear only twice a year: If the asset has a missing report, we remove them from the universe. We do not consider that they have the same report as a quarter ago and then use the same report. We also rearranged the temporal arrangement of features constructed from fundamental data, so that information appears only the moment they become available to the public, or *point-in-time* (PIT) adjusted. We address this challenge with thorough attention, data leakage can invalidate our whole procedure of comparison.

Balance sheet currencies are converted from other currencies to NOK using currency pairs from Yahoo finance.

Since data is reported quarterly, and we would like to align balance sheet generated factors to factors with daily frequencies when we estimate factor returns, we utilize the imputation technique of last-observation-carried-forward. The method relies on imputing the last observed values, until a new and true observation reveals itself. These values are updated around every business quarter, which is roughly 63 days assuming 252 trading days in a year, whenever PIT-adjusted financial report data presents itself.

It is worth noting that measurement errors in the variables present themselves when we use LOCF-based imputation. A fundamental variable observed at report date is realistically assumed to be noiseless. In practice, the fundamentals or balance sheet data-generating processes of firms are more easily believable, to be latent variables which evolve beyond what we observe. Our imputation is therefore just a proxy of the latent variable, and it is rational to believe that the signal-to-noise ratio degrades as days pass from when the observation was truly updated by a financial report.

Outlier detection and excessive correlation: Excessive correlation means that the dataset is given with a complete history of a company with all the tickers that the company has reported under. If a company has changed tickers they report under, the first ticker will still be visible using the same data as the new ticker. To remove these double reporting data, a correlation model was run through the data to make sure that none of the companies reported twice. Tickers with a correlation coefficient $\rho > 90\%$ required further inspection.

After merging old tickers with new symbol names, removing duplicates, and removing tickers with illegitimate data, our universe shrunk from 526 to 340 tickers.

For the sake of simplicity, we construct the idiosyncratic volatility of firms from a 60-day rolling CAPM regression with a minimum of 20 observations.

The statistical significance of mean factor returns has been checked using t-statistics.

3.5 Reliability/credibility of the investigation

3.5.1 Look-ahead bias

Adjusted close prices carry information about future performance of a financial instrument, with dividend, buybacks, or split adjustments reported at later points in time, look-ahead bias may therefore present itself if price reflects a synthetically low or high value (Paleologo, 2025). Adjusted close price returns do not carry information about the future, because they do not carry. In addition, they reflect the investors true wealth gain, in contrast to unadjusted close prices. When using stock price data in feature construction, we resorted to using the unadjusted close price. For features which utilized return data, we have used the adjusted close price returns.

3.5.2 Data sources

EODHD does not record brent oil futures prices or offer currency pairs in our subscribed package; As mentioned, Yahoo Finance was used as a data provider. Data integrity may therefore be compromised due to the use of data from unique vendors. In a few cases some extreme return observations were made, which upon personal was often confirmed to be illegitimate price movements, due to the failure of adjusting for more complex splits and capital restructuring on the Oslo stock exchange. A few valid tickers also lacked balance sheet data, which led to the decision to drop them as entities as they had no available data for factor construction. We assume that EODHD provides correct and legitimate data, with the amount of split errors found in outlier detection and the lack of report dates and proper merging of old to new tickers, the quality of our sourced data poses a serious threat to the integrity of the study.

3.5.3 Changes in the code after producing results

Once an Out-of-sample test is executed and results are viewed, there are very few reasons as to why adjusting any part within the model construction procedure, and run the test again for new results is well reasoned. Changing and fixing models after OOS-results are prone to being forged results, and we treat it very cautiously. We did change a few parts of the model pipeline, and we believe it is barely justified because it can be viewed as

bug-fixing and not as ex-post model tuning;

1. We discovered that the smoothing parameter in the exponential weighting of the characteristic model idiosyncratic covariance matrix were misspecified. Whereas common convention has the smoothing parameter as $\theta = 0.95$ as the persistence, the simple exponential smoothing function from the forecast-package in R, has the convention defining the smoothing parameter to decide how much, to weigh the most recent observation. During the first OOS-test, the characteristic idiosyncratic covariance matrix had a smoothing parameter with a value of 0.05 using if we followed common convention like risk-metrics (Alexander, 2008), which means that only the most recent observation were given weight. We set the function smoothing parameter as $1 - \theta$, and after application, both the characteristic and statistical factor models have an exponential weighting scheme that is identical to each other.
2. We increase the walk-forward OOS forecasting window and adding a naive sample covariance benchmark. This was both, to evaluate both more advanced factor models in comparison to simple estimation method, in addition to evaluating how the different models perform under various market regimes. The walk-forward out-of-sample size was increased from spanning 2024 to 2025, to spanning 2020-2025, this effectively assigns the characteristic model a smaller sample size for estimation. However, both the statistical factor model and naive sample covariance is limited to their pre-defined rolling sample window consisting of 252 trading days. In the earlier version, we compared only the characteristic and statistical factor. Moreover, we believed that the naive covariance can provide nuance to the comparison. We will disclose our previous test results of the characteristic and statistical model using the OOS period from 2024 to 2025 in the results section.

3.6 Validity/transferability of the investigation

3.6.1 Apples to apples, or apples to pears?

Both models share unique abilities and similar traits. Our aim with comparison was to build the models from scratch, aiming for choosing model choices that draw both models towards an equal footing. For example, the models are quite different up until the point of choosing the weighting scheme for the covariance matrix, where both models receive an exponential weighting with a specified smoothing parameter that yields an identical half-life. By weighting scheme, we showcase both forecasting methods advantages, and on the surface both are similar to generalized least squares. The main difference being that we assume that idiosyncratic volatility is proportional to the inverse of firm market capitalization for the characteristic factor model, and specify the weighting matrix in the cross-sectional WLS-regressions thereby. While the less data-intensive statistical model

whitens the return matrix by reducing idiosyncratic noise, and reveal richer correlation structure for improved factor estimation.

In retrospect we doubt that the convention of adding a dynamic choosing mechanism for determining the number of factors for the statistical model was fair choice, and in practice we believe that one would expect to see results drop if a static number of factors were chosen throughout the test sample. However, the characteristic model has variable exposures dependent on micro-economic traits, which establish factor returns that should convey a dynamic story of the covariance between firms. In this sense, they are both dynamic.

3.6.2 Model and test assumptions

Both the characteristic model and statistical model require a handful of assumptions to be present for them to be considered an efficient estimator of covariance in our study. For the characteristic model, weighted linear regression assumptions apply for achieving unbiased and efficient estimation of factor returns. We assume that our specified exogenous variables are linear in parameters. How realistic is this? Most variables, value, size, momentum, reversal, and betas/sensitivities are seemingly linear characteristics. We find the linearity assumption easier to challenge in variables such as CAPM estimated idiosyncratic volatility, yet the purpose of the natural log transformation is supposed to form a more symmetric distribution of volatility exposures. Even more doubtful are dividend yields and illiquidity. Illiquidity might have been a more well-behaved measure if there were no firms who had no trading activity for a single day. To get to the point, a fraction with a zero in the denominator is undefined. However, we believe that a 252-day rolling measure may at least counteract these “no-trading activity” observations, in contrast to the original measure introduced by Amihud (REF). Dividend yield suffers a unique issue, firms sharing dividends is often regarded as positive by market participants. However, in some circumstances, 12-month trailing dividend yields can reach above 100%, indicating that the given stock has suffered a significant fall in its price level, often signaling distress. To improve this, one could construct step-wise dividend yield features in a regression model; setting a threshold for the level of dividend yield, where being under the threshold relates to positive market signals, and being over the threshold relates to firm distress. However, to effectively carry out our main research objective of this master’s thesis, we sought to keep predictive modeling parsimonious.

4 Reproduceability and transferability

The extract-transform-load data pipeline was not uploaded to the appendix as it was an excessively large collection of various feature engineering programs. We do plan on

creating a GitHub repository and upload all code after the evaluation of this thesis is completed. And with that, one should be able to reproduce our results by buying the *all-in-one* commercial data license on EODHD, assuming EODHD does not post-process the related data.

4.0.1 LLM Disclaimer

We utilized Large Language Models for conversational aid for learning and reflecting on included literature, in addition to exploring and searching existing relevant literature, generating preliminary programming drafts (Python, R and LaTeX) and bug fixing, especially concerning API calls from our data provider. All code and programming drafts have been carefully reviewed and extensively modified. Large Language Models were not used to generate text, formulate scientific claims, make methodological decisions, or analytical conclusions. Reasoning, methodology and analysis, as well as written content within this thesis, are the authors own work.

5 Results and Discussion

5.1 Factor Returns

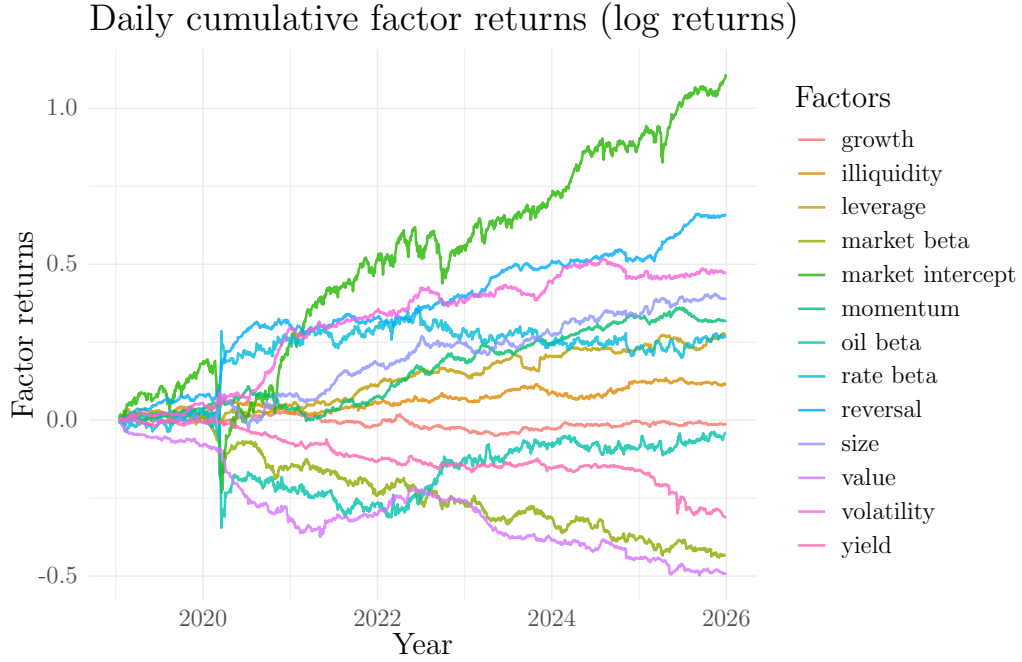


Figure 2. A multi series plot of cumulative logarithmic returns of the various risk factors included in the characteristic model.

Here is a plot of several of the cumulative return paths realized by the factors. The returns are transformed to log-returns, so there is only a visual bias for symmetry and visual appeal's sake. Log-returns have not been used to create features, or return-series used in estimation. The multi-plot, serves as visual guide for exploring the relationships variables might have. Seemingly, the movements several factors are not obviously correlated. However, economic distress events, especially COVID-19, are apparent and a significant correlation can be observed relating to the event can be observed.

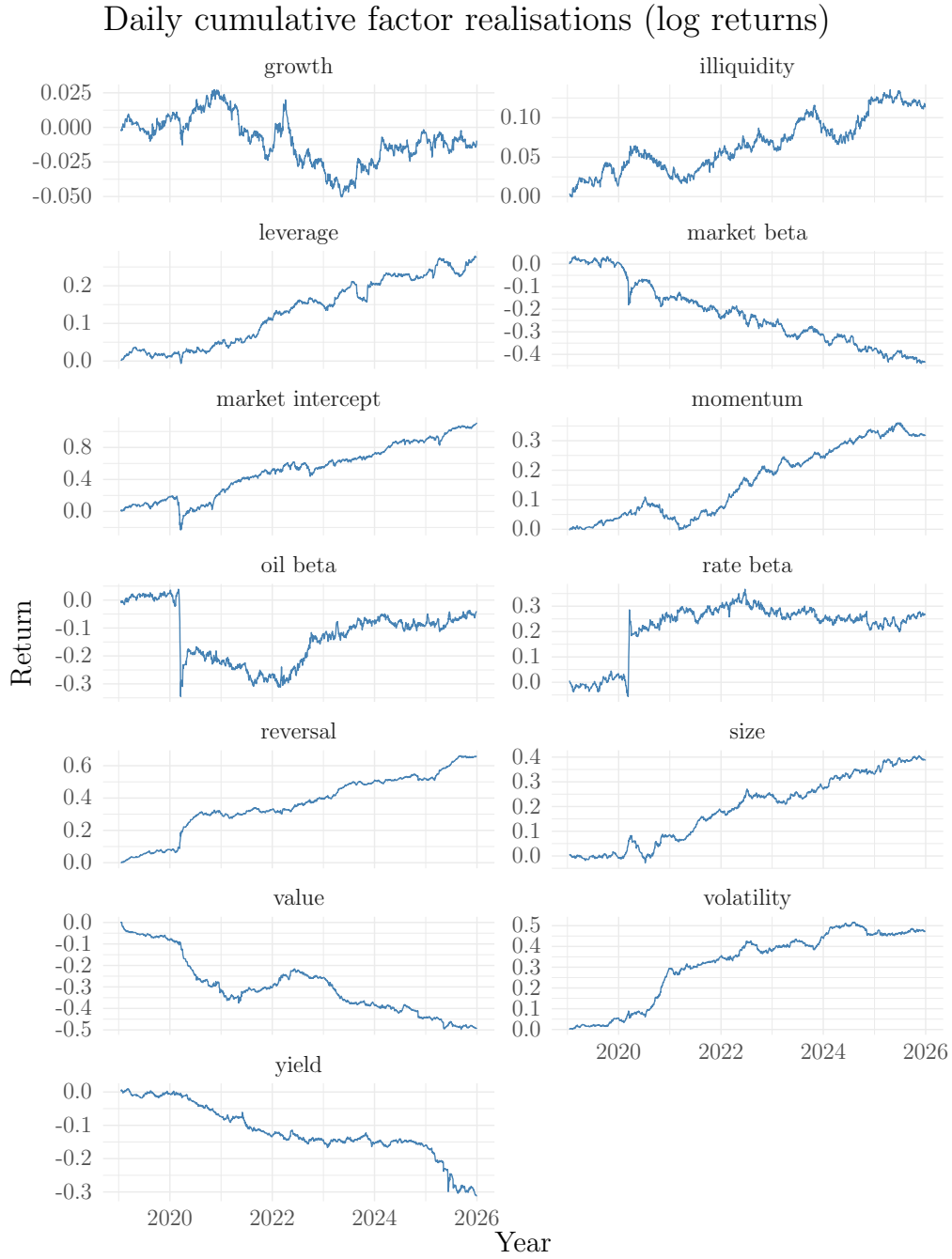


Figure 3. A multi series plot of cumulative logarithmic returns of the various risk factors included in the characteristic model.

We also present the cumulative returns of the factors in their respective plots. The reason we follow up on visual interpretation, is to diagnose or identify any misspecification that may occur. These are not asset premia in the common academic sense; they should, however, reflect how the documented anomaly behaves, but this does not essentially mean the that the. The returns can be interpreted as the return on the factor with exposure equal to one. A common specification is to note that value and size are heading opposite directions, which coincides with Fama and French (1993) findings that their value and size

on portfolio sorts are inversely correlated. However, they are both pointing in opposite directions to what the common literature remarks.

Table 2. ADF-test, testing the stationarity of the estimated factor returns.

	t-statistic	p-value
(Intercept)	-40.889	0.01
value	-39.369	0.01
size	-39.845	0.01
momentum	-40.719	0.01
reversal	-39.295	0.01
volatility	-40.631	0.01
illiquidity	-46.681	0.01
yield	-48.070	0.01
oil_beta	-34.696	0.01
mkt_beta	-45.820	0.01
rate_beta	-33.478	0.01
leverage	-41.844	0.01
growth	-41.614	0.01

5.2 Model Diagnostics

5.2.1 Characteristic Factor Model

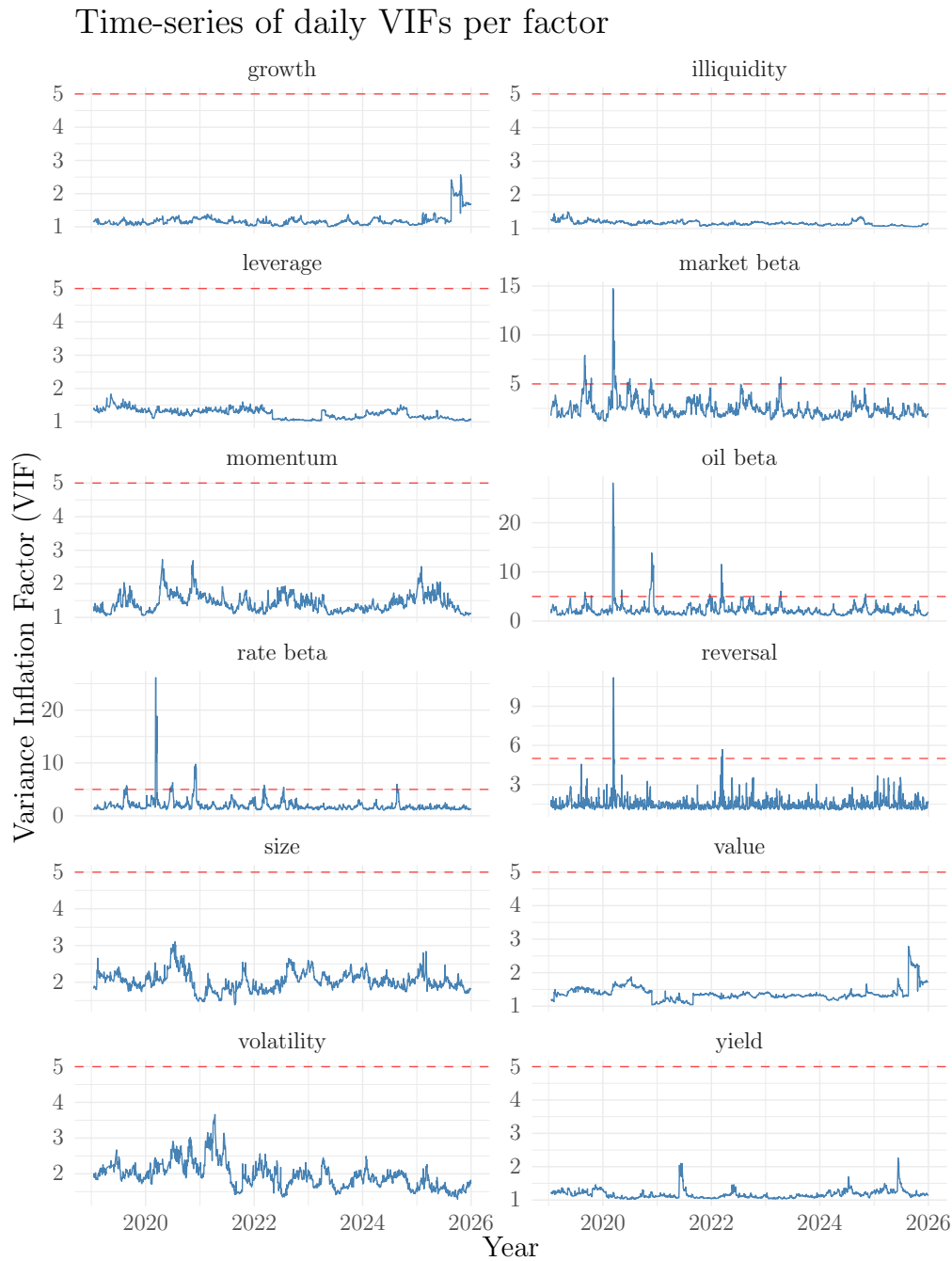


Figure 4. A time-series of *variance-inflation* (VIF) values of each cross-section for each characteristic. $VIF < 5$ reflects low multicollinearity, $5 < VIF < 10$ reflects moderate to high multicollinearity, and $10 < VIF$ reflects severe multicollinearity among regressors. The red dotted line is set at a value of 5

Here we observe a plot of the daily estimated variance inflation (VIF) for all factors used in estimation. The VIF diagnostic aids in diagnosing how prominent multicollinearity increases variance in estimates. Even though factor return estimates are unbiased, a widely

known heuristic is that a VIF value above 5 indicates troublesome multicollinearity and renders estimates more unstable and uncertain. The technical term is variance inducing, troublesome multicollinearity increases the standard error of the factor return estimates. From the time-series plot of daily VIFs, we observe a few problematic spikes, but we note that the VIFs are relatively stable most of the time, and most often in normal geopolitical regimes. The spikes we observe are most likely and seemingly connected to geopolitical events which cause systematic instability among financial markets. We can visually draw the connection, to events such as the events of COVID-19 (March 2020) and the Russia-Ukraine invasion (February 2022) to name a few. We can interpret this through two nuances. While it is convenient and a desired result that multicollinearity remains stable and rarely induces excessive variance in estimation, it is unfortunate to know that the factor returns estimates are most unstable when a risk manager needs reliable estimates the most, particularly in unstable financial market regimes.

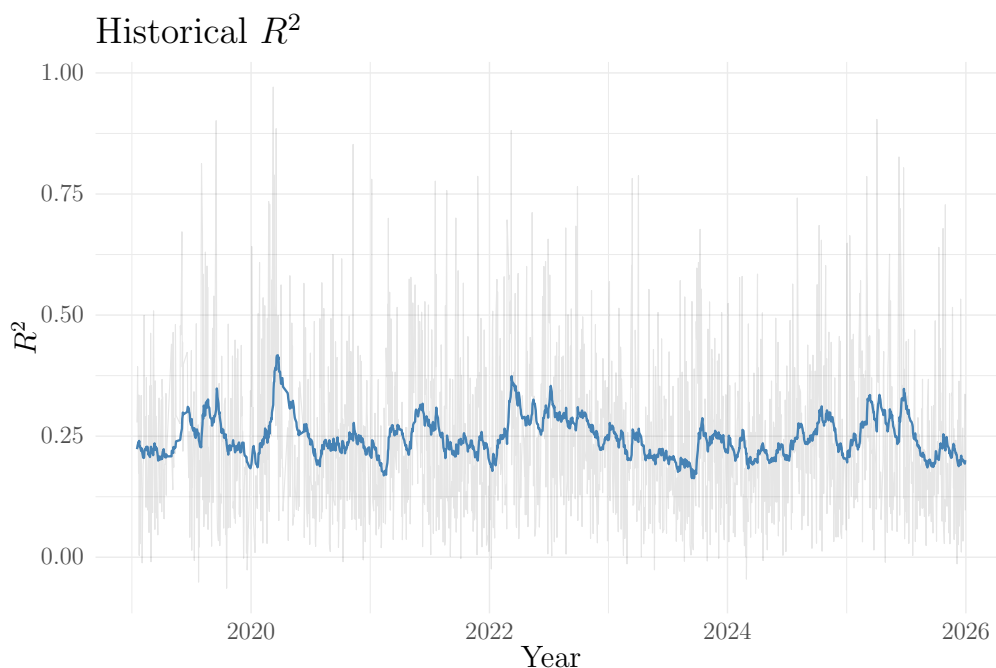


Figure 5. A plot of historically estimated adjusted R^2 in faint gray which reflect the in-sample explanatory power of each cross-sectional regression. The blue line is a smoothed average of the R^2 .

We can also observe the historical adjusted R^2 , above with a smoothed line in the middle indicating its trending value across cross sections. We also provide a description of the distributions of the values R^2 throughout the historical cross-sections.

Table 3. Descriptive statistics of adjusted R^2 across cross-sections.

	Adj. R^2
Max	0.9709
Q3	0.3369
Median	0.2199
Mean	0.2469
Q1	0.1221
Min	-0.0640

We can observe that it is quite volatile with significant variation across cross-sections.

The cross-sectional characteristics spike and degrade spuriously, seemingly explaining almost all of the variation in returns on a given day, to almost none in another. In the few cases, where correlation is quite high, *e.g.* $R^2 > 0.75$ the model likely captures crisis regimes or high asset correlation events that occur. In the latter case, we are left with little idiosyncratic noise among firms, which in turn may underestimate risk forecasts when estimated. Grinold and Kahn (2007) notes that larger R^2 values tend to appear in observations of larger market moves. Our estimation procedure may also be subject to microstructure noise, we chose a daily estimation frequency for factor returns, but it is worth noting that the BARRA commercial models estimate factor returns monthly (Grinold & Kahn, 2007). There are other metrics that are appealing to us, such as the median and mean R^2 . At the time, Grinold and Kahn (2007) remarks that a commercial BARRA model, used to estimate 50 various factor returns achieves an average R^2 between 30 to 40 percent given monthly estimation on 1000 assets.

5.2.2 Statistical Factor Model

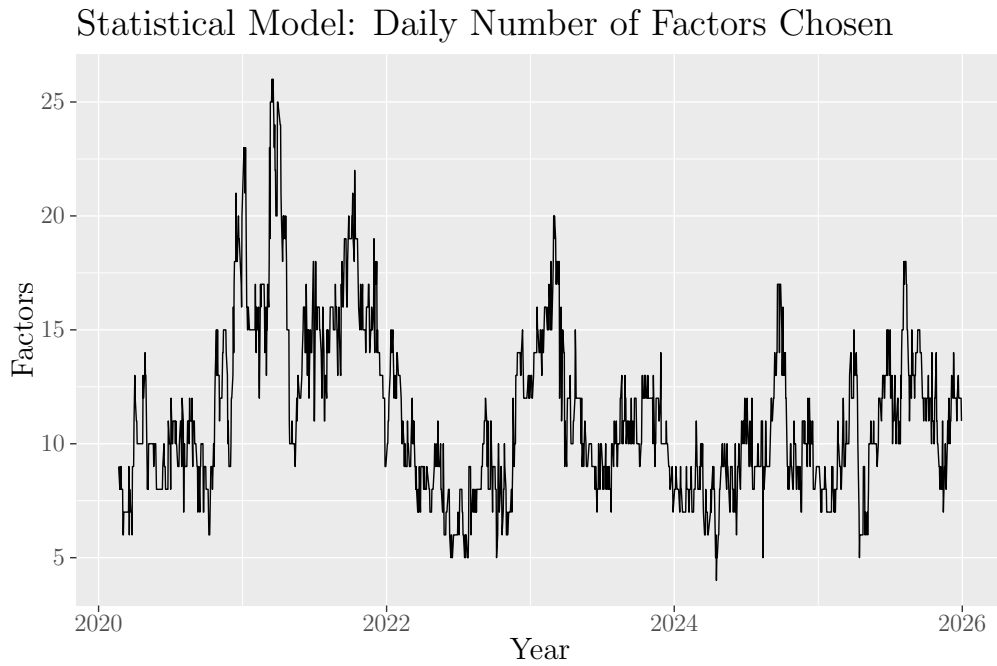


Figure 6. A time-series plot of the statistical factor model’s dynamic choice of the number of useful factors.

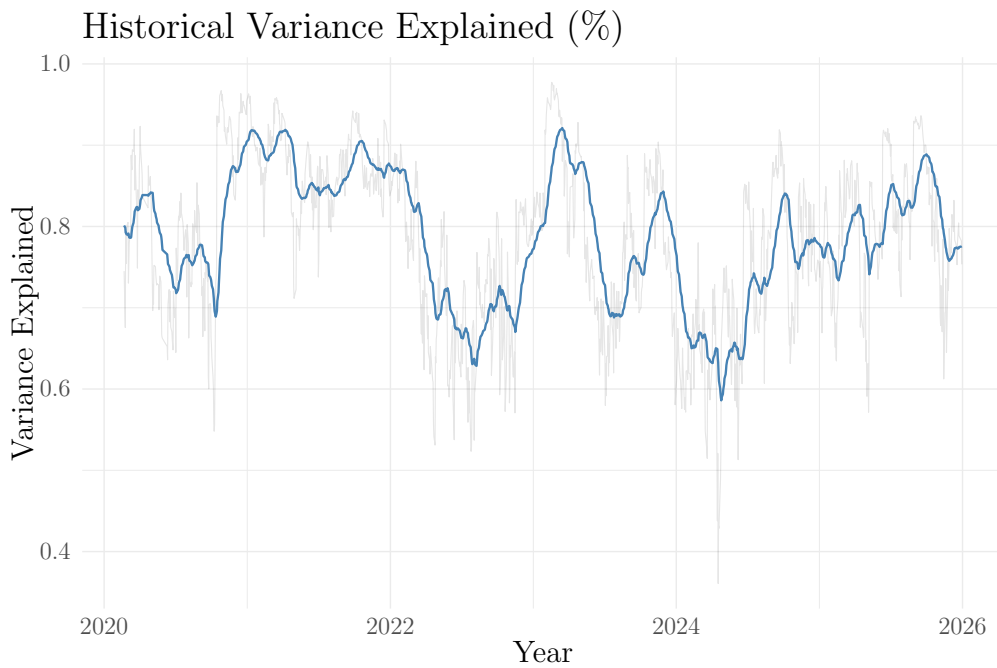


Figure 7. A time-series plot of the statistical factor model in-sample variance explained for 252-day backwards looking sample window.

We observe Figure 6 and 7, which move unsurprisingly synchronized. The more factors that are utilized by the information-criteria dynamic selection method, the more variance

the statistical factor model explains. In relation to the smoothed realized variance graphs discussed later in this section, we observe that the model is selecting the largest number of factors on average compared to other periods in Figure 6. It is also worth noting that the statistical factor model requires a smaller number of factors both at the beginning of 2020 and at the end 2025 to maintain a significant proportion of variance explained.

5.3 Factor selection and noise

5.4 Test Results and Portfolio Statistics

Table 4. Diebold-Mariano test of expanding $h = 1$ forecast.

Comparison	DM statistic	p-value
Characteristic vs. Naive	-0.982	0.326
Statistical vs. Naive	-1.645	0.100
Characteristic vs. Statistical	0.458	0.647

We should also account for multiple testing to reduce the chance of receiving a false-positive. There are a number of subjective decisions involved in factor model construction, some of which we are subject to when choosing winsorization methods, outlier removal, half-life conditional sample sizes, *etc.* López De Prado (2023) demonstrates that the financial factor investing literature is subject to p-hacking, and is even more compounded by publication bias. Therefore, we choose a confidence level of 99%.

A negative DM-statistic would favor the first mentioned model. For example, in the comparison of the Characteristic-based covariance and naive sample covariance, the DM-statistic is negative indicating that the loss differentials are in favor of the Characteristic model. However, the test finds no significant difference in loss-differentials, implied by the $p\text{-value} = 0.326$, which is not even near a 99%-confidence level. The difference in loss-differentials is greater for the statistical-based covariance in comparison to the naive sample covariance. This test also finds no significant difference between the statistical-based covariance and the naive sample covariance. When we compare the characteristic and statistical-based covariance, the DM-statistic of loss differentials favors the statistical model. However, the difference in loss differentials was also found to be insignificant for the two main models, with a $p\text{-value} = 0.647$. We have previously remarked an earlier version of the walk-forward OOS GMV portfolios, disclosing in the validity section. The test only concerned comparing the characteristic and statistical factor model.

Table 5. Yearly annualized volatility of GMV portfolios/ratios in comparison to the characteristic model.

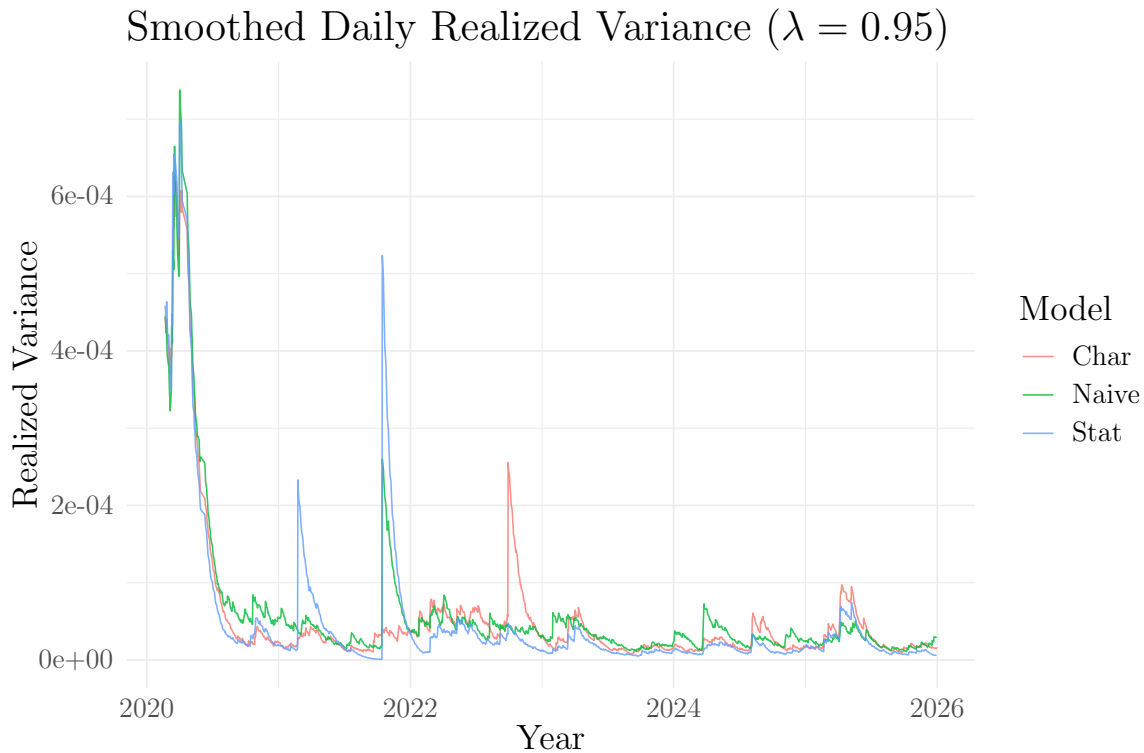
Year	$\sigma_{Characteristic}$	$\sigma_{Statistical}$	σ_{Naive}	$\frac{Statistical}{Characteristic}$	$\frac{Naive}{Characteristic}$
2020	0.178	0.175	0.197	0.983	1.105
2021	0.080	0.132	0.106	1.650	1.334
2022	0.126	0.082	0.107	0.649	0.844
2023	0.074	0.060	0.084	0.813	1.138
2024	0.073	0.058	0.084	0.797	1.151
2025	0.087	0.071	0.078	0.821	0.898

Table 6. Yearly average GMV portfolio leverage

Year	Characteristic	Statistical	Naive
2020	1.709	1.473	3.226
2021	1.500	1.192	2.380
2022	1.805	1.276	2.790
2023	1.515	1.199	2.782
2024	1.474	1.160	2.790
2025	1.546	1.229	2.983

Using the portfolio statistics for diagnostics, we make no causal claims, but exclaim the differences portfolio statistics across the model results. In Table ??, we can observe the different annualized volatility each model portfolio has realized for each year. The characteristic-based covariance performs best of all covariance matrices in 2021, but yields ≈ 0.107 in annualized volatility across the whole forecasting period, while the Naive sample covariance has the worst annualized volatility for the whole period at ≈ 0.113 , and the statistical model has the best annualized volatility across the whole period at ≈ 0.102 . The statistical model performs best from 2022-2025, yielding the smallest OOS annualized volatilities for the GMV portfolio for each year. The latter can also be viewed in terms ratios of annualized volatilities each year where $\frac{Statistical}{Characteristic} > 1$ indicating that the characteristic GMV portfolio performed better. The ratio also confirms that the statistical-based GMV portfolio performed better in 2022 to the end of 2025. Another interesting metric we have included is leverage. The GMV portfolio solution presented earlier is unconstrained, and the weights generated by the solution is not restricted to entering short positions, which is why the fully invested GMV portfolio has leverage. Leverage is calculated as the sum of the absolute value of the portfolio weights $Leverage = \sum_i |w_i|$. A high value of leverage conveys that the GMV portfolio

requires entering a handful of short positions in stocks to achieve the optimal GMV portfolio. From the table 10 showing the yearly average leverage applied to form the three respective portfolios from each covariance matrix, we observe a clear ranking. The naive sample covariance require significant leverage to form the GMV portfolio. The statistical GMV portfolio requires the least amount of leverage in order to form the GMV portfolio. These results are valuable for fund manager who wish to construct similar portfolios. Although not impossible, the GM portfolio generated by the statistical factor model is less trivial to trade. Due to rotational indeterminacy, loadings are only identified up to an arbitrary orthogonal rotation, and the turnover on statistical factor portfolios is high if no engineering is applied (Paleologo, 2025). Moreover, it is more trivial to trade the characteristic-based portfolios due to the rotational determinacy. However, the leverage required is still non-trivial.



We have smoothed the plots for visual appeal. The smoothing parameter can only convey the weighted average realized variance. In practice, the realized variance spikes are instantaneous, and short lasting. Its main use is to make significant spikes visually identifiable and separable across models. We analyze the spikes chronologically. We can observe that all models spike in volatility in the extraordinary global pandemic event COVID-19. The Naive sample covariance GMV portfolio uses more time to recover from the pandemic. We observe that the statistical GMV portfolio spikes in early 2021, significantly in contrast to the characteristic and naive GMV portfolios. The next spike that occurs in the end of 2022 is intriguing. Both the naive and statistical GMV portfolios spike in realized variance, while the characteristic GMV portfolio remains relatively stable, before

all three model portfolios experience increased variance for the remainder of 2022. At the end of 2022 the characteristic GMV portfolio spikes in light of the other two GMV portfolios. An explanation, may be spurious effects created by the oil beta factor, due to the fact that brent oil futures rise significantly in to the middle of 2022. However, the latter implies that the effect of the oil beta is lagged, which is doubtful. The volatility spikes observed in all models calms with only small spikes appearing from 2023 to the end of 2025.

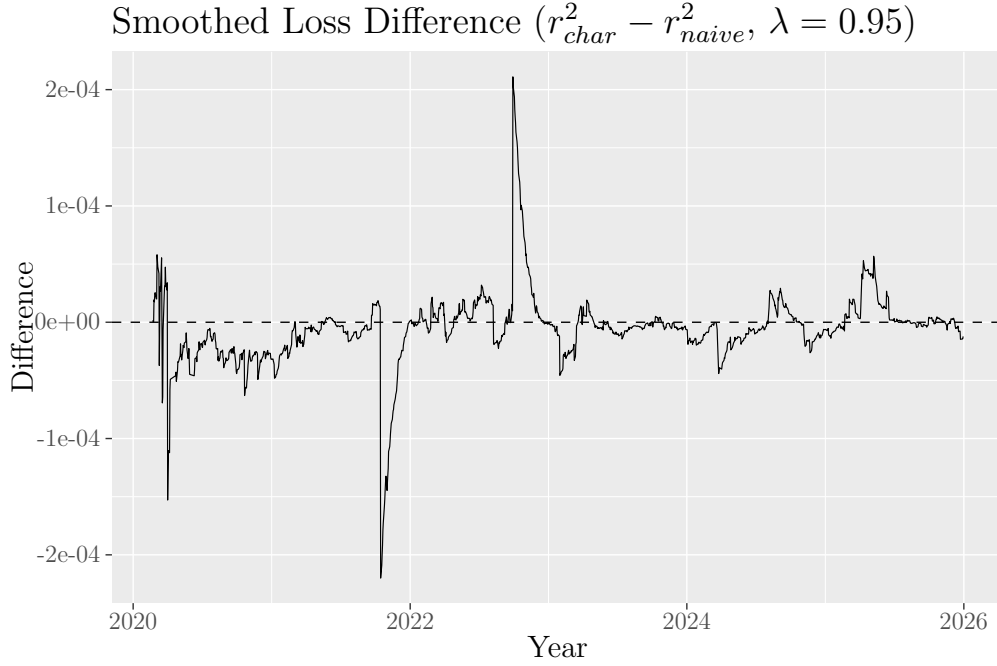


Figure 8. A smoothed difference plot of the realized variances of the GMV portfolios generated by the characteristic factor model and pairwise observation estimated sample covariance.

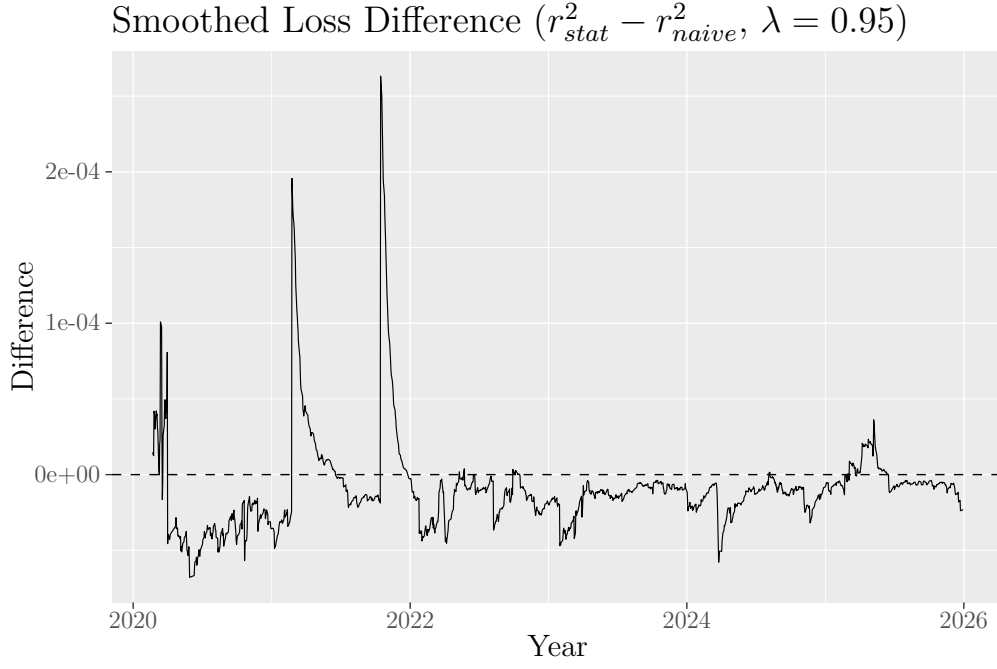


Figure 9. A smoothed difference plot of the realized variances of the GMV portfolios generated by the statistical factor model and pairwise observation estimated sample covariance.

5.4.1 Characteristic vs. Statistical

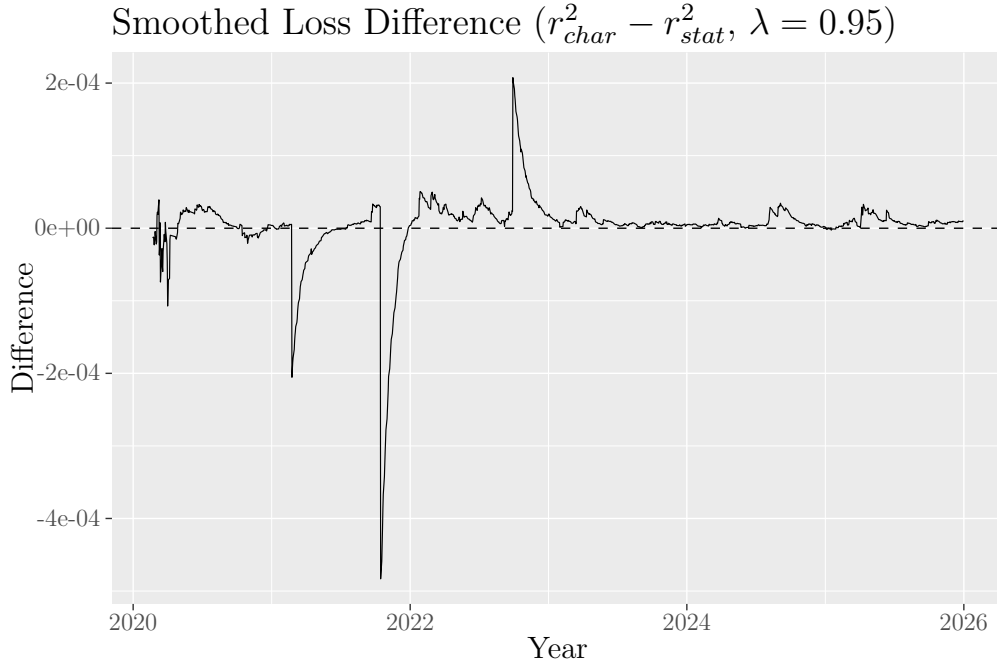


Figure 10. A smoothed difference plot of the realized variances of the GMV portfolios generated by the characteristic factor model and the statistical factor model.

6 Conclusion

In this thesis, we have compared the out-of-sample walk-forward test to answer the question of whether the characteristic-based or statistical-based covariance derived from their respective factor models are better at predicting portfolio risk on the Oslo stock exchange. We also included the naive sample covariance to serve as minimum benchmark.

We find no significant differences in the performance of the OOS GMV portfolios realized variance loss differentials through Diebold-Mariano tests. From the portfolio statistics in table 9, we observe that statistical factor model had the lowest annualized volatility for the longest periods compared to the other models, but the difference was not found statistically significant. Surprisingly, we find in our tests that the naive sample covariance generated GMV portfolios has no significant difference in performance compared to the other two models. We observed that the statistical model demanded the least leverage, and the naive sample covariance demanded a larger amount of leverage. This means that which model selected for risk estimation comes down to other preferences. Both of the models have their strengths, the characteristic model is more data-intensive, but is more interpretable than the statistical model.

We exclaim that the lack of significance in the DM-test does not imply that there exist no difference between the models. The ADF-stationarity test came out significant rejecting the unit root, but in retrospect, we have realized that exponential smoothing estimators implicitly assume local stationarity of covariance, allowing it to change according to our set smoothing parameter's half-life. The DM-test's assumption of covariance stationarity, is directly broken because of our modeling choices of letting covariance vary across time, and therefore we can not benefit from the asymptotic properties the test offers. The models may also be subject to misspecification. Systematic sources of risk that are documented in the US literature does not imply that they are directly transferable to the Norwegian equity market. We note that all the intuition based factors we have constructed and included, such as the yield and oil beta provide some spurious regression in market stress regimes.

As mentioned, the quality of our data is questionable, however we saw to process it and maintain as best as we can. We tried controlling for survivorship bias, but may only have controlled for lesser proportion of it due to the lack of and presence of faulty. Our characteristic factor model, feature construction and estimation is subject to various forms of measurement error in LOCF imputed fundamentals and variables with estimation objects such as betas, used to construct them, in addition to being prone to microstructure noise. Barra models at the time of Grinold and Kahn (2007), were estimated monthly. Paleologo (2025) exclaims that when the factor covariance is estimated on shorter frequencies, autocorrelation may become more pronounced. As mentioned covariance stationarity is

a challenged assumption, our loss differentials carry, being subject to several extreme regimes through time such as COVID-19 and Russia-Ukraine invasion.

We propose several improvements to both models and testing frameworks. One the use of Kalman filters for both the characteristic and statistical factor model. The Kalman filter is an adaptive Bayesian-like algorithm that accounts for measurement noise and autocorrelation. One can also find it useful to model the persistence of correlation and volatility dynamics separately (Engle, 2002). Unfortunately, we discovered the testing framework of Raffaella Giacomini and White (2006) at the end of our thesis timeline, which is an extension of the Diebold-Mariano test. Raffaella Giacomini and White (2006). Their testing framework favors models with rolling or dynamic weighting of observations, or in other words forecasting methods that use local data. They introduce the concept of the conditional null hypothesis, where a regime specific indicator can be added (e.g. VIX, SOFR, *etc.*), and models can be tested under various regimes. The null hypothesis compares the loss differentials similar to the DM-test conditioned on a regime indicator. The null hypothesis states that both models, and today's regime indicator can not predict which model is better. We believe that using this testing framework instead would be significant improvement to the Diebold-Mariano test. We could not perform it, as all our candidate regime indicators (OSE returns and NOWA), were used in feature construction.

Our thesis results are limited, but we contributed to the literature by providing a thorough and guided estimation process.

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Portfolio risk forecasting has been and continues to be an active research field for both academics and practitioners. Almost all institutional investment management firms use quantitative models for their portfolio forecasting, and researchers have explored models' econometric foundations, relative performance, and implications for capital market behavior and asset pricing equilibrium. Portfolio Risk Analysis provides an insightful and thorough overview of financial risk modeling, with an emphasis on practical applications, empirical reality, and historical perspective. Beginning with mean-variance analysis and the capital asset pricing model, the authors give a comprehensive and detailed account of factor models, which are the key to successful risk analysis in every economic climate. Topics range from the

relative merits of fundamental, statistical, and macroeconomic models, to GARCH and other time series models, to the properties of the VIX volatility index. The book covers both mainstream and alternative asset classes, and includes in-depth treatments of model integration and evaluation. Credit and liquidity risk and the uncertainty of extreme events are examined in an intuitive and rigorous way. An extensive literature review accompanies each topic. The authors complement basic modeling techniques with references to applications, empirical studies, and advanced mathematical texts. This book is essential for financial practitioners, researchers, scholars, and students who want to understand the nature of financial markets or work toward improving them

Cover – Title – Copyright – Contents – Acknowledgments – Introduction – Key Notation – 1 Measures of Risk and Return – 1.1 Measuring Return – 1.2 The Key Portfolio Risk Measures – 1.3 Risk-Return Preferences and Portfolio Optimization – 1.4 The Capital Asset Pricing Model and Its Applications to Risk Analysis – 1.5 The Objectives and Limitations of Portfolio Risk Analysis – 2 Unstructured Covariance Matrices – 2.1 Estimating Return Covariance Matrices – 2.2 The Error-Maximization Problem – 2.3 Portfolio Choice as Decision Making under Uncertainty – 3 Industry and Country Risk – 3.1 Industry-Country Component Models – 3.2 Empirical Evidence on the Relative Magnitudes of Country and Industry Risks – 3.3 Sector-Currency Models of Corporate Bond Returns – 4 Statistical Factor Analysis – 4.1 Types of Factor Models – 4.2 Approximate Factor Models – 4.3 The Arbitrage Pricing Theory – 4.4 Small-n Estimation Methods – 4.5 Large-n Estimation Methods – 4.6 Number of Factors – 5 The Macroeconomy and Portfolio Risk – 5.1 Estimating Macroeconomic Factor Models – 5.2 Event Studies of Macroeconomic Announcements – 5.3 Macroeconomic Policy Endogeneity – 5.4 Business Cycle Betas – 5.5 Empirical Fit and the Relative Value of Macroeconomic Factor Models – 6 Security Characteristics and Pervasive Risk Factors – 6.1 Equity and Fixed-Income Characteristics – 6.2 Characteristic-Based Factor Models of Equities – 6.3 The Fama-French Model and Extensions – 6.4 The Semiparametric Approach to Characteristic-Based Factor Models – 7 Measuring and Hedging Foreign Exchange Risk – 7.1 Definitions of Foreign Exchange Risk – 7.2 Optimal Currency Hedging – 7.3 Currency Covariances with Stock and Bond Returns – 7.4 Macroeconomic Influences on Currency Returns – 8 Integrated Risk Models – 8.1 Global and Regional Integration Trends.

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Appendices

A Statistical model estimation

The *singular* value decomposition $R = USV^\top$ of asset returns will be extensively used to achieve the spiked covariance model. The aim of the spiked covariance model is to recover and distinguish a latent factor structure⁴ to improve the signal-to-noise ratio of the covariance matrix of asset returns. Similar to the characteristic factor model we assume idiosyncratic volatilities are uncorrelated, which yields us the strict spiked covariance model. Most of the procedure is adapted from Paleologo (2025), but it is worth noting that Connor et al. (2010) presents the exact model without an exponential weighting scheme, but refers to the scalar factor model.

1. **Inputs:** $R \in \mathbb{R}^{n \times T}$. Assets are represented in rows, and dates are column wise. Using this convention is necessary so that the SVD of R returns $U \in \mathbb{R}^{n \times n}$ representing the cross-sectional dimension, and $V \in \mathbb{R}^{T \times T}$ representing the time dimension. The convention will become apparent when we employ weighting schemes.
2. It is practical to model the persistence of correlation and volatility dynamics differently. In our paper, we did not give the exponential weighting scheme for estimating the characteristic factor model covariance, different half-lives for correlation, and volatility. We will therefore also not assign this ability to the statistical factor model. We can create an exponentially weighted return matrix by setting

$$W_\alpha := c \times \text{diag}(e^{-(T/\alpha)}, \dots, e^{-(1/\alpha)})$$

$$R_\alpha = RW_\alpha$$

. Where c is a normalizing constant, used to make the sum of squared diagonal weights equal to T .

3. Perform an SVD/PCA on the exponentially weighted returns R_α . Boivin and Ng (2006) propose a weighting scheme similar to generalized least squares for improving factor model estimates, where we select a factors ex-ante to estimate the idiosyncratic volatilities, with the aim of making the return data more homoscedastic. Here

⁴Given that it is identifiable.

Boivin and Ng (2006) leans towards selecting more factors rather than few to adequately estimate idiosyncratic standard deviations. Bai and Ng (2002) claims that information-criterion based factor selection is more efficient in identifying the true number of factors, when heterogeneous idiosyncratic noise in the data is reduced. Building on the latter proposition, when the noise is reduced, the data should be exclusively noisy or exhibit a systematic factor structure.

4. From the above item, we define the idiosyncratic weighting matrix W_σ as a diagonal matrix of inverse idiosyncratic or residual volatilities estimated from the previous factor model, whose sole purpose is to estimate residuals. The return matrix in the final stage is weighted by both W_σ and W_α , yielding $\bar{R} = W_\sigma R W_\alpha$.
5. The whitening process allows us to move closer towards fulfilling the assumptions of the probability PCA of Tipping and Bishop (1999) of isotropic noise, which in econometric language means the simultaneous assumption of homoskedasticity and uncorrelated errors.
6. At last, it is important to note, that the spiked covariance must span the original space by de-weighting loadings and adding back the idiosyncratic noise we estimated and de-weighted earlier, so that it is not underestimated. We can set the factor covariance to $\Sigma_f := (\Lambda_k - \hat{\lambda})$, where Λ_k is a diagonal matrix of the k -selected factors by Bai and Ng (2002) IC3. Our final model which can be viewed in the code is presented as follows

$$\Sigma_R = B \Sigma_f B^\top + \hat{\sigma}^2 \hat{\lambda}$$

. Where $\hat{\sigma}^2$ is the estimated idiosyncratic variances from whitening factor model.

None of the literature proposes how to define the parameter α so that the half-life is equal to that of the EWMA used in our characteristic factor model. To make the weighting schemes between exponential weighting matrix and the EWMA equivalent. We need the half-life of the matrix to be properly defined by requiring $0.5 = e^{-(\text{half-life}_\theta/\alpha)}$. Our set smoothing parameter is $\theta = 0.95$, and our half-life is

$$\theta^{\text{half-life}} = \frac{1}{2} \implies \frac{\ln(0.5)}{\ln(\theta)} = \text{half-life}$$

. This yields ≈ 13.51 , and then,

$$0.5 = e^{-(13.51/\alpha)} \implies \alpha \approx \frac{13.51}{\ln(2)}$$

. Even, though the constant c scales the weighting matrix its purpose is to scale eigenvalues, and the calibration beforehand was to show that it is possible to achieve equal decay in weighting recent observations with both the weighting matrix and EWMA.

B Sandwich Bias-correction

Paleologo (2025) remarks that the estimated factor covariance is inflated $\hat{\Sigma}_f = \Sigma_f + (B^\top \Sigma_\epsilon^{-1} B)^{-1}$, and can be bias corrected. $\bar{\Sigma}_f = \hat{\Sigma}_f - (B^\top \Sigma_\epsilon^{-1} B)^{-1}$. Notice that the bias $(B^\top \Sigma_\epsilon^{-1} B)^{-1}$, is the variance of the estimated coefficient or factor return $\text{var}(\hat{f})$ from a regression, this connects to a key result from econometrics. For example, from Davidson and MacKinnon (2004) call this form a sandwich covariance matrix. We can therefore also correct for using our imperfect proxy $W_m = \text{diag}(\sqrt{mcap_i}, \dots, \sqrt{mcap_N})$. Our factor returns are estimated through the WLS,

$$\hat{f} = (B^\top W B)^{-1} B^\top W R$$

we can derive the sandwich correction,

$$\hat{f} = (B^\top W B)^{-1} B^\top W (Bf + \epsilon)$$

$$\hat{f} - f = (B^\top W B)^{-1} B^\top W \epsilon$$

$$\text{var}(\hat{f}) = \mathbb{E}[(\hat{f} - f)(\hat{f} - f)^\top]$$

$$\text{var}(\hat{f}) = (B^\top W B)^{-1} B^\top W \epsilon \epsilon^\top W B (B^\top W B)^{-1}$$

$$\text{var}(\hat{f}) = (B^\top W B)^{-1} B^\top W \Sigma_\epsilon W B (B^\top W B)^{-1}$$

When if it is true that $W = \Sigma_\epsilon^{-1}$, the correction term, or sandwich collapses,

$$(B^\top \Sigma_\epsilon^{-1} B)^{-1} B^\top \Sigma_\epsilon^{-1} \Sigma_\epsilon \Sigma_\epsilon^{-1} B (B^\top \Sigma_\epsilon^{-1} B)^{-1}$$

.

$$\implies (B^\top \Sigma_\epsilon^{-1} B)^{-1}$$

.

(Davidson & MacKinnon, 2004, p. 197).

C Tables

C.1 A taxonomy of factors

Table 7. Taxonomy of Financial Factors in Risk-Factor Modeling

Factor	Proxy	References
Value (Book-to-Market)	$\frac{\text{Book Equity}_{i,t}}{\text{Market Cap}_{i,t}}$	Daniel and Titman (1997) and Fama and French (1993)
Size	$\ln(\text{Market Cap}_{i,t})$	Fama and French (1993) and Van Dijk (2011)
Momentum	$\prod_{d=t-252}^{t-21} (1 + r_{i,d}) - 1$	Carhart (1997) and Jegadeesh and Titman (1993)
Weekly Reversal	$-\sum_{d=t-5}^{t-1} r_{i,d}$	Lehmann (1990)
Log of Idio. Vol.	$\ln(\text{Var}(\hat{\varepsilon}_{i,t}^m))$	Ang et al. (2006)
Rolling Illiquidity	$\frac{1}{K} \sum_{d=t-K+1}^t \frac{ R_{i,d} }{\text{NOK Volume}_{i,d}}$	Amihud (2002)
Dividend Yield	$\frac{\sum_{d \in [t-252, t]} \text{Dividends}_{i,d}}{\text{Price}_{i,t}}$	Naranjo et al. (1998) and Zimmer (2024)
Oil Beta	$\hat{\beta}_i^{\text{Oil}} = \frac{\text{Cov}(r_{i,t}, r_{\text{Oil},t})}{\text{Var}(r_{\text{Oil},t})}$	Chen et al. (1986)
Market Beta	$\hat{\beta}_i^m = \frac{\text{Cov}(r_{i,t}, r_{m,t})}{\text{Var}(r_{m,t})}$	Frazzini and Pedersen (2014) and Sharpe (1964)
Rate Sensitivity	$\hat{\beta}_i^y = \frac{\text{Cov}(r_{i,t}, r_{y,t})}{\text{Var}(r_{y,t})}$	Chen et al. (1986)
Growth	$\frac{\Delta \text{Total Assets}_{i,t}}{\text{Total Assets}_{i,t-1}}$	Cooper et al. (2008)

C.2 R^2 statistics across cross-sections

Table 8. Descriptive statistics of adjusted R^2 across cross-sections.

	Adj. R^2
Max	0.9709
Q3	0.3369
Median	0.2199
Mean	0.2469
Q1	0.1221
Min	-0.0640

C.3 Portfolio statistics

Table 9. Yearly annualized volatility of GMV portfolios/ratios in comparison to the characteristic model.

Year	$\sigma_{Characteristic}$	$\sigma_{Statistical}$	σ_{Naive}	$\frac{Statistical}{Characteristic}$	$\frac{Naive}{Characteristic}$
2020	0.178	0.175	0.197	0.983	1.105
2021	0.080	0.132	0.106	1.650	1.334
2022	0.126	0.082	0.107	0.649	0.844
2023	0.074	0.060	0.084	0.813	1.138
2024	0.073	0.058	0.084	0.797	1.151
2025	0.087	0.071	0.078	0.821	0.898

Table 10. Yearly average GMV portfolio leverage

Year	Characteristic	Statistical	Naive
2020	1.709	1.473	3.226
2021	1.500	1.192	2.380
2022	1.805	1.276	2.790
2023	1.515	1.199	2.782
2024	1.474	1.160	2.790
2025	1.546	1.229	2.983

C.4 Diebold-Mariano test

Table 11. Diebold-Mariano test of expanding $h = 1$ forecast.

Comparison	DM statistic	p-value
Characteristic vs. Naive	-0.982	0.326
Statistical vs. Naive	-1.645	0.100
Characteristic vs. Statistical	0.458	0.647

D Figures

D.1 Experimental design

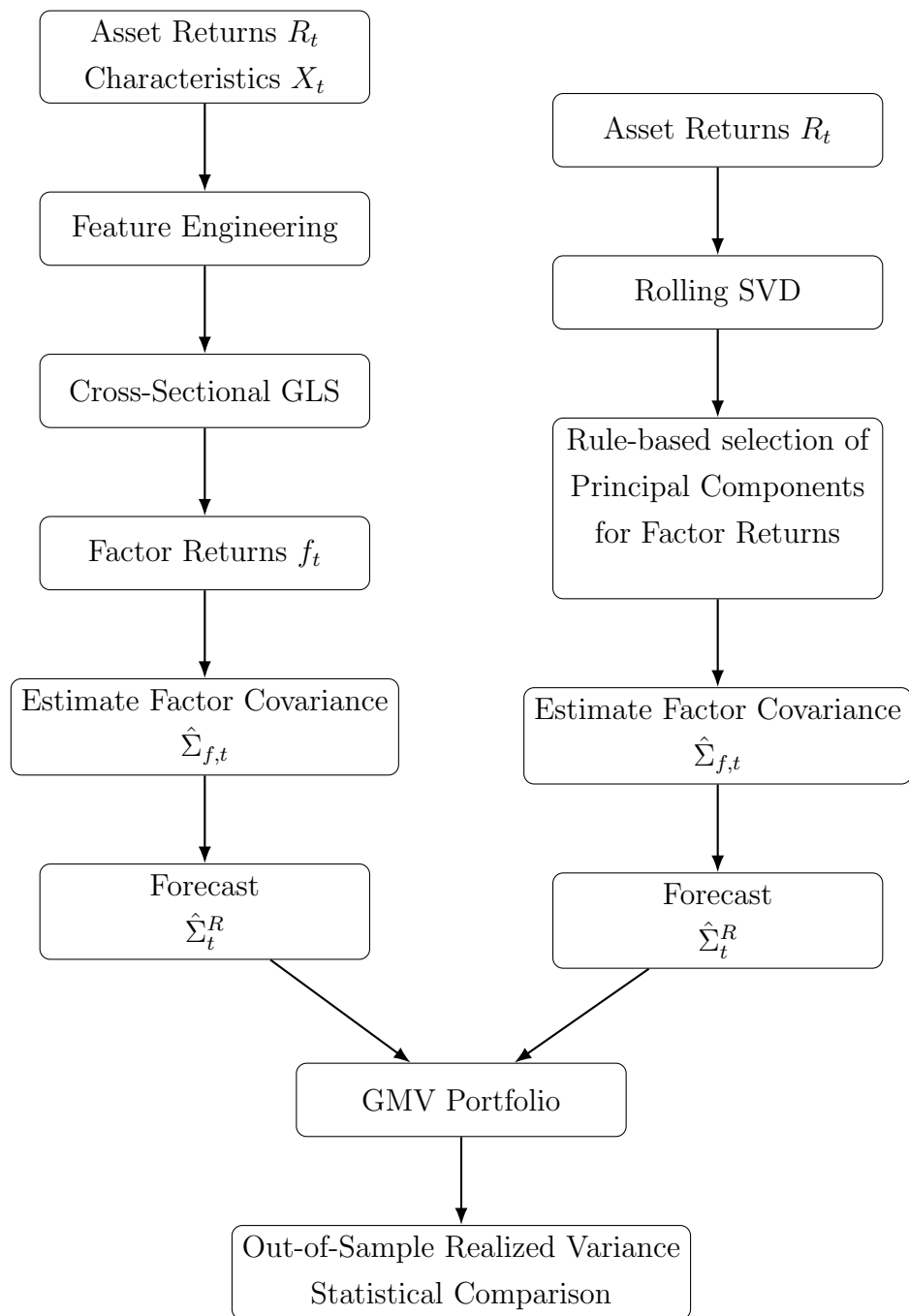


Figure 11. Experimental Design for Model Construction and Comparison.

D.2 Factor returns

D.2.1 Multi plot

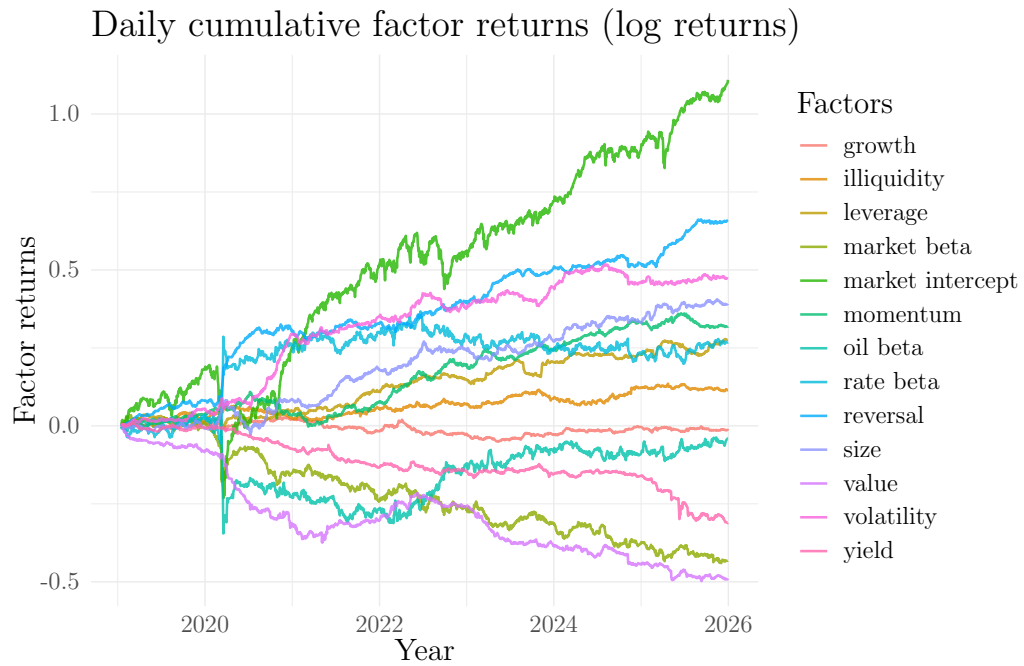


Figure 12. A multi series plot of cumulative logarithmic returns of the various risk factors included in the characteristic model.

D.2.2 Single plots

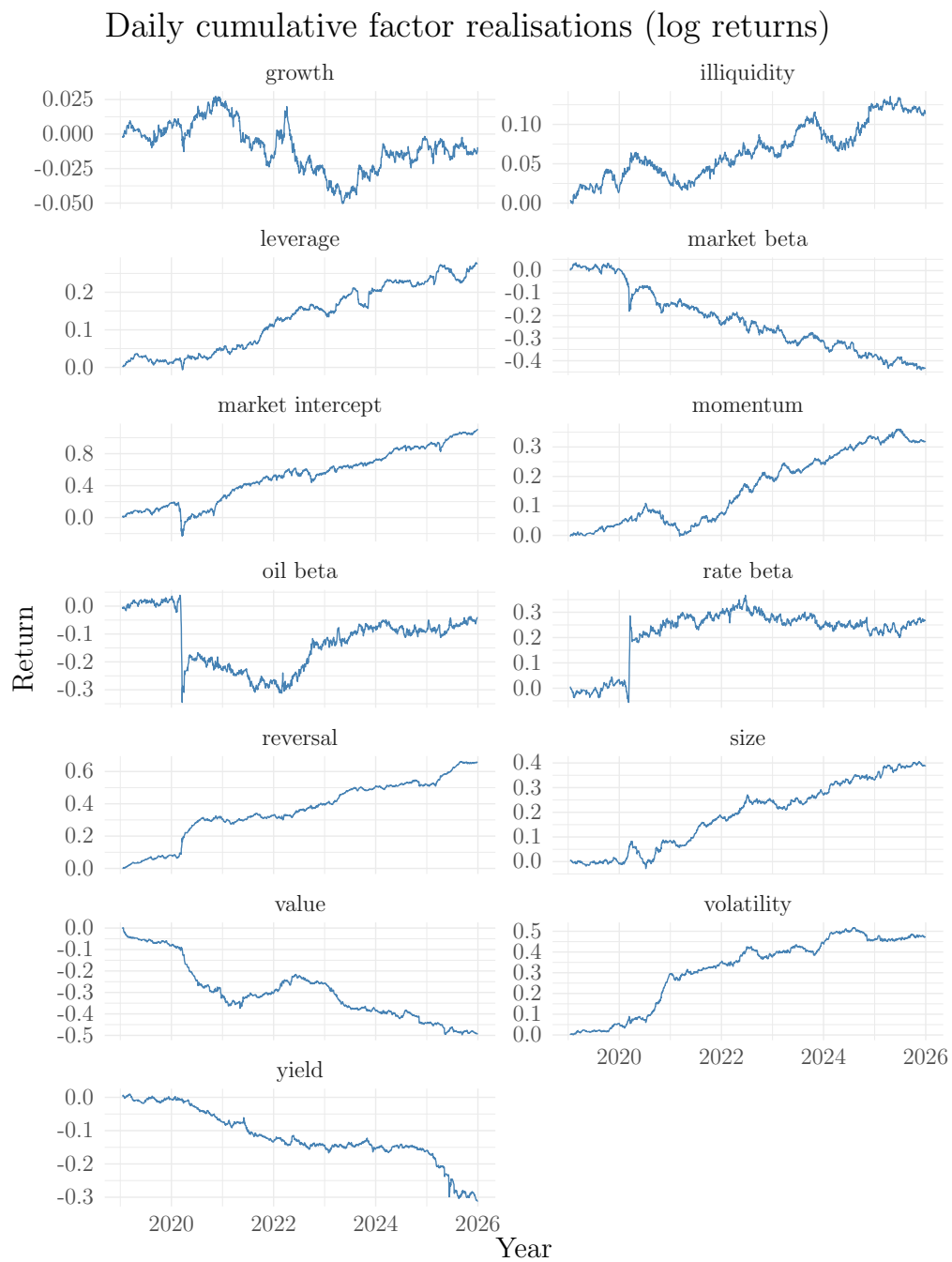
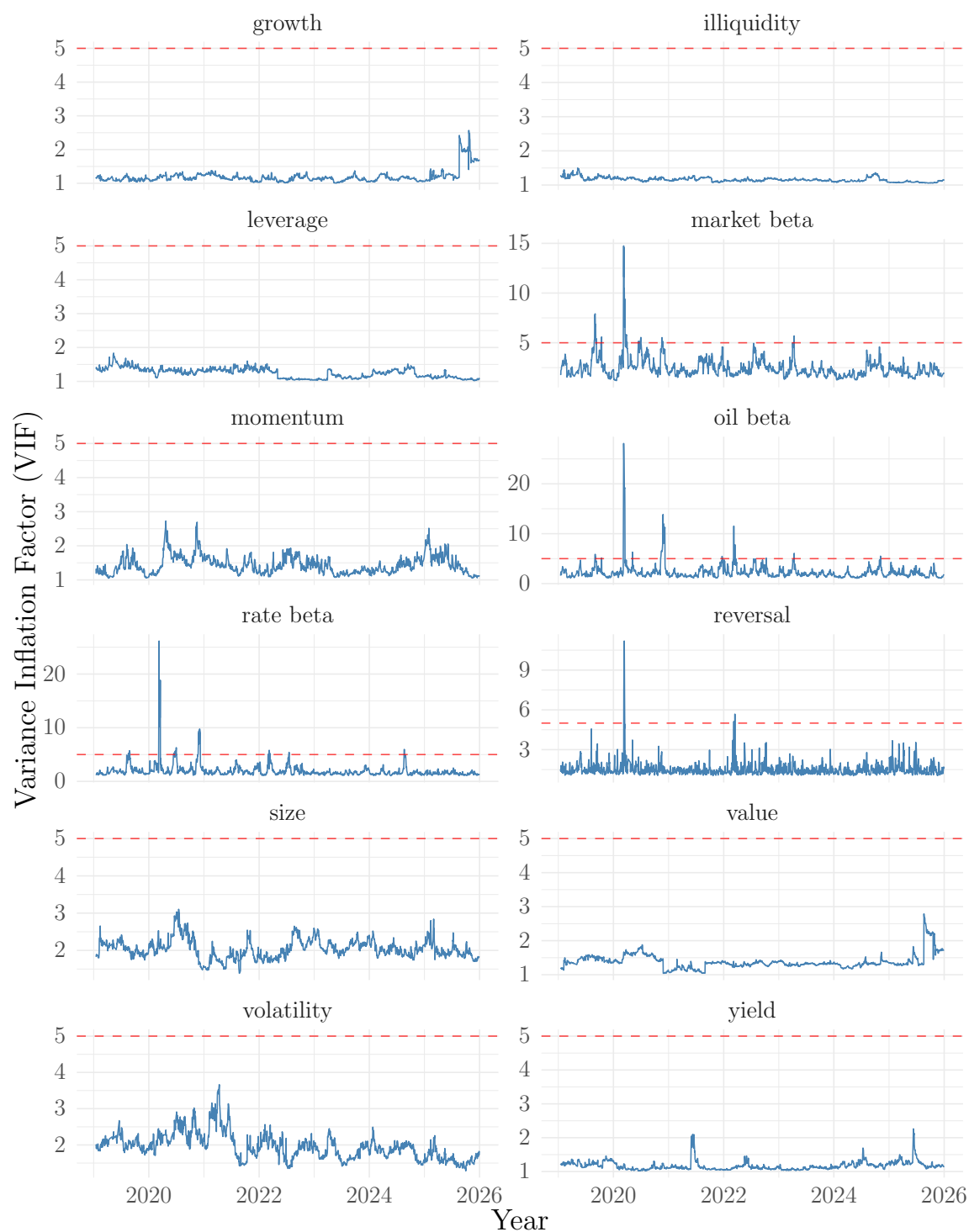


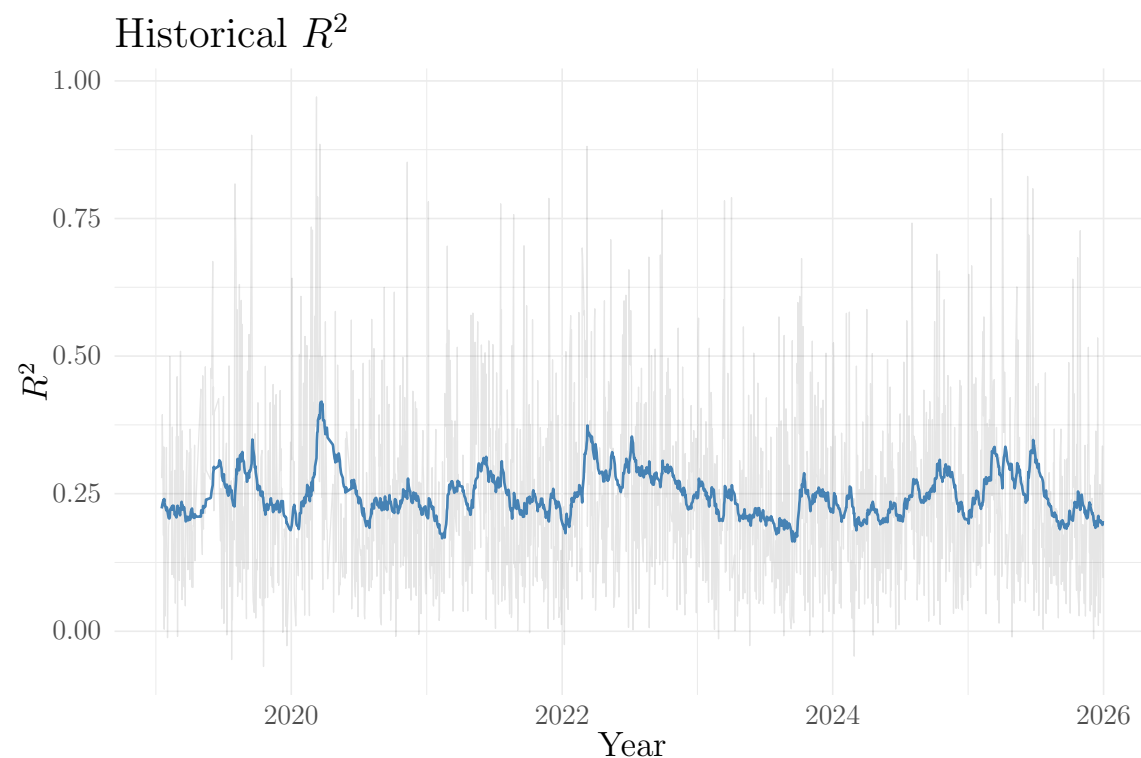
Figure 13. A multi series plot of cumulative logarithmic returns of the various risk factors included in the characteristic model.

D.3 Historical VIF across cross-sections

Time-series of daily VIFs per factor



D.4 Historical R^2 across cross-sections



D.5 Smoothed realized variance difference plots

D.5.1 Characteristic vs. Naive

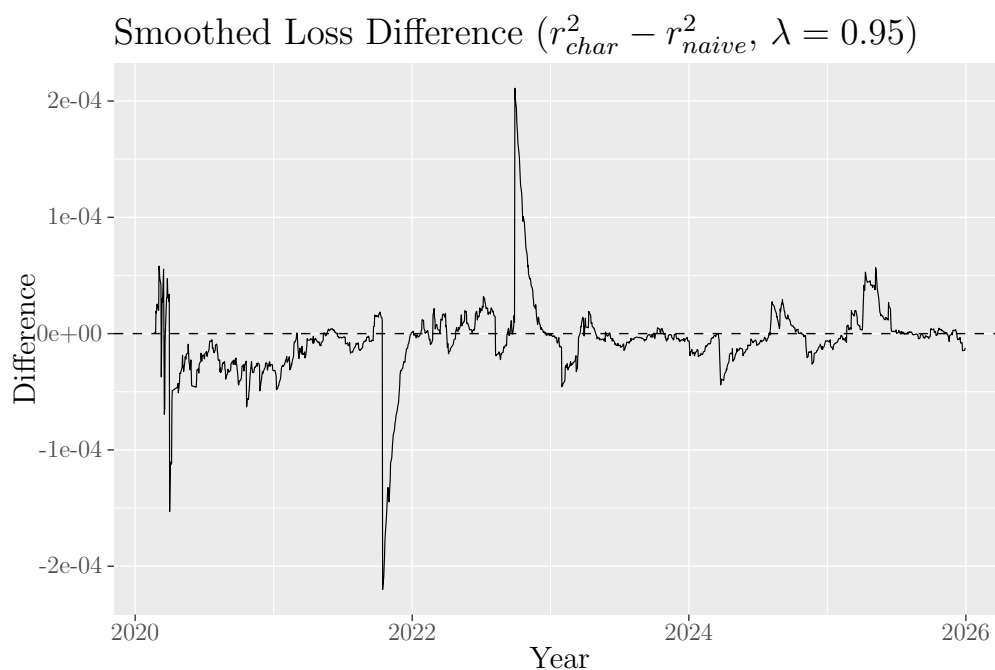


Figure 14. A smoothed difference plot of the realized variances of the GMV portfolios generated by the characteristic factor model and pairwise observation estimated sample covariance.

T

D.5.2 Statistical vs. Naive

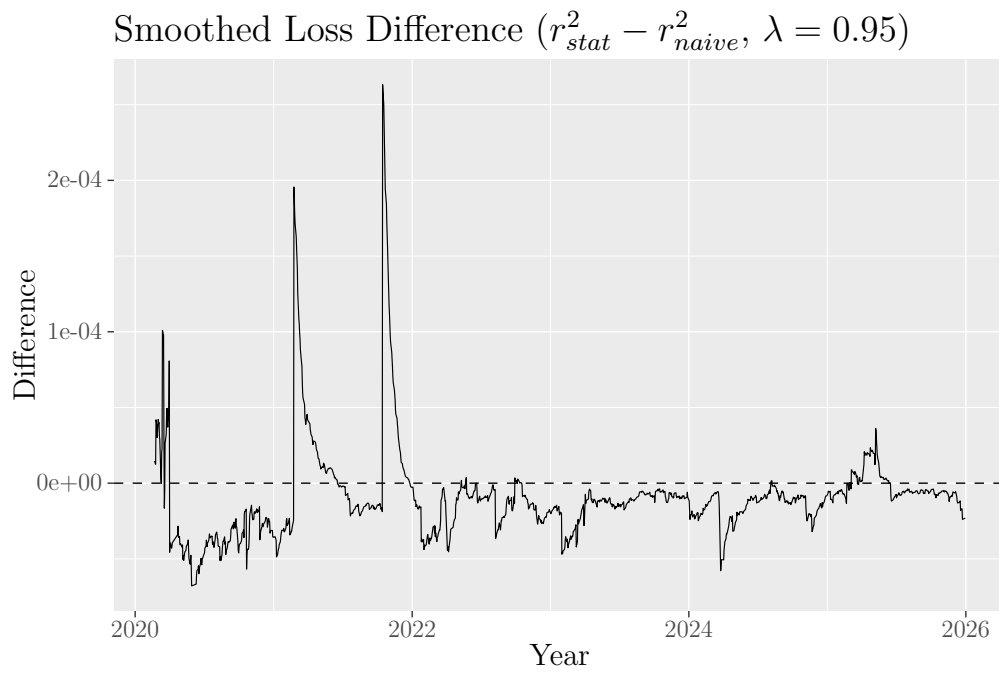
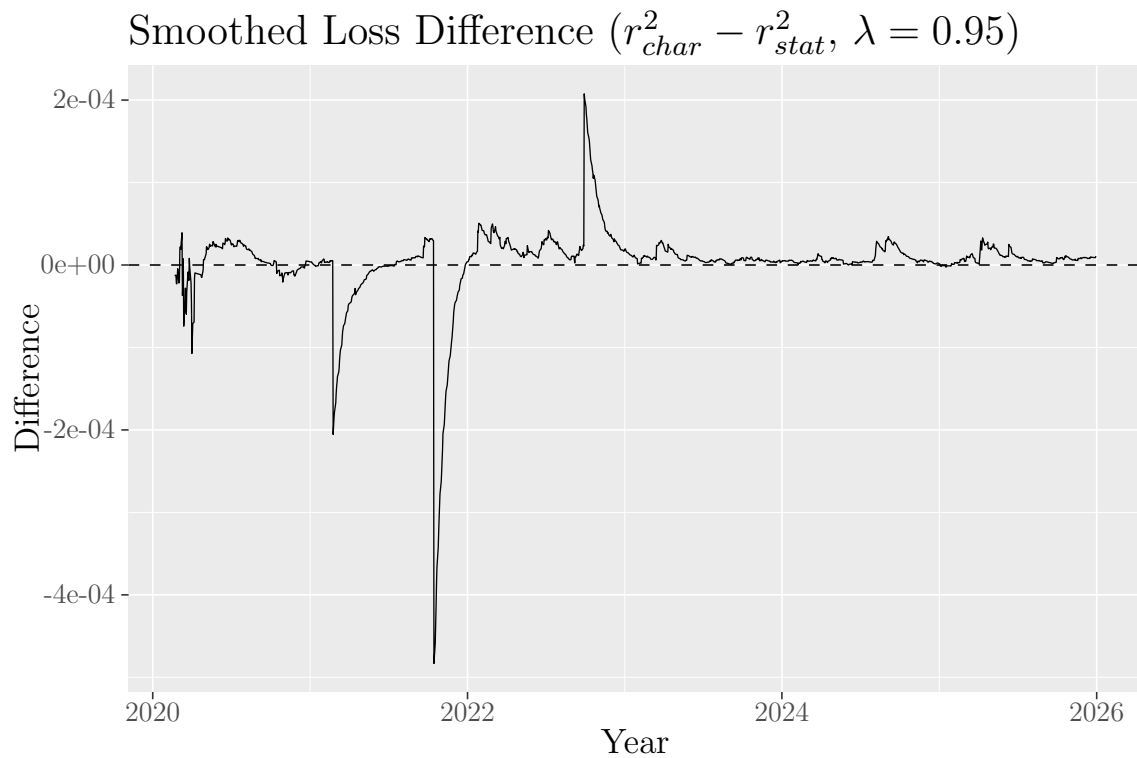


Figure 15. A smoothed difference plot of the realized variances of the GMV portfolios generated by the statistical factor model and pairwise observation estimated sample covariance.

D.5.3 Characteristic vs. Statistical



D.6 R-Code

```
1
2 # Initialize session
3 ```{r}
4 knitr::opts_chunk$set(echo = TRUE)
5 rm(list = ls())
6 setwd("Set work directory of choice")
7 ```
8
9 # Libraries
10
11 ```{r, include = FALSE}
12 libs <- c("tidyr", "tidyverse", "dplyr", "lubridate", "ggplot2", "lmtest",
13   ↪ "PerformanceAnalytics", "forecast", "corrplot", "regclass", "pcaMethods",
14   ↪ "tseries", "tikzDevice", "knitr", "sandwich")
15
16 for (i in libs) {
17   library(package=i, character.only=TRUE)
18 }
19
20 # options for generating LaTeX plots
21 options(tikzLatex = "~/pdflatex.exe",
22   tikzDefaultEngine = "pdftex",
23   sanitize = TRUE,
24   tikzLatexPackages = c(
25     getOption("tikzLatexPackages"),
26     "\\usepackage{amsmath}",
27     "\\usepackage{amssymb}",
28     "\\usepackage{bm}"
29   ))
30
31 tikzDevice::tikzTest()
32
33 # LaTeX plot converter function
34 tex_plot <- function(plot, filename = "plot.tex", width = 6, height = 4) {
35   tikz(filename,
36     width = width,
37     height = height,
38     standAlone = FALSE)
39
40   print(plot)
41
42   dev.off()
43 }
44
45 # Generate well formatted summary statistics
```

```

44 gen_desc_statistics <- function(df, decimals = 4) {
45   sapply(as.data.frame(df), function(x) {
46     c(
47       Max      = round(max(x), decimals),
48       Q3       = round(quantile(x, 0.75, names = FALSE), decimals),
49       Median   = round(median(x, 0.5), decimals),
50       Mean     = round(mean(x), decimals),
51       Q1       = round(quantile(x, 0.25, names = FALSE), decimals),
52       Min      = round(min(x), decimals)
53     )
54   })
55 }
56 ```
57
58
59
60 # Load data
61
62 ```{r}
63 # Price series
64 # adjusted close prices (dates x tickers)
65 adj_close <- read.csv("final_adj_close_prices_aligned.csv", row.names = 1) %>%
66   ↪ as.data.frame()
67 rownames(adj_close) <- as.Date(rownames(adj_close))
68
69 # calculate returns
70 returns <- (adj_close / lag(adj_close)) - 1
71
72 osebx <- read.csv("xosl_prices.csv", row.names = 1) %>% as.data.frame()
73 osebx_adj_close <- osebx$adjusted_close %>% as.data.frame()
74 rownames(osebx_adj_close) <- as.Date(rownames(osebx))
75
76 osebx_returns <- (osebx_adj_close / lag(osebx_adj_close)) - 1
77
78 # Time-series of BM-ratios (dates x tickers)
79 value_ts <- read.csv("book-to-market(value)_aligned_final.csv", row.names = 1) %>%
80   ↪ as.data.frame()
81 rownames(value_ts) <- as.Date(rownames(value_ts))
82
83 # Time-series of the logarithms of market capitalizations (dates x tickers)
84 size_ts <- read.csv("log_mcap(size)_aligned_final.csv", row.names = 1) %>%
85   ↪ as.data.frame()
86 rownames(size_ts) <- as.Date(rownames(size_ts))
87
88 # Time-series of historical relative momentum (dates x tickers)
89 momentum_ts <- read.csv("momentum_aligned_final.csv", row.names = 1) %>%
90   ↪ as.data.frame()

```

```

87 rownames(momentum_ts) <- as.Date(rownames(momentum_ts))
88
89 # Time-series of historical short-term reversal (dates x tickers)
90 reversal_ts <- read.csv("weekly_reversal_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
91 rownames(reversal_ts) <- as.Date(rownames(reversal_ts))
92
93 # Time-series of logarithm 60-day rolling CAPM idiosyncratic volatilities (dates x
  ↪ tickers)
94 volatility_ts <- read.csv("log_idio_vol_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
95 rownames(volatility_ts) <- as.Date(rownames(volatility_ts))
96
97 # Time-series of 12-month rolling Amihud illiquidity measure (dates x tickers)
98 illiquidity_ts <- read.csv("illiquidity_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
99 rownames(illiquidity_ts) <- as.Date(rownames(illiquidity_ts))
100
101 # Time-series of 12-month rolling dividend yields (dates x tickers), 0 for tickers who
  ↪ does not pay dividends.
102 yield_ts <- read.csv("div_yield_aligned_final.csv", row.names = 1) %>% as.data.frame()
103 rownames(yield_ts) <- as.Date(rownames(yield_ts))
104
105 # Time-series of 21-day rolling Brent Oil Futures beta (dates x tickers)
106 oil.beta_ts <- read.csv("oil_rolling_betas_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
107 rownames(oil.beta_ts) <- as.Date(rownames(oil.beta_ts))
108
109 # Time-series of 60-day rolling asset to market beta (dates x tickers)
110 market.beta_ts <- read.csv("market_beta_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
111 rownames(market.beta_ts) <- as.Date(rownames(market.beta_ts))
112
113 # Time-series of 21-day rolling Brent Oil Futures beta (dates x tickers)
114 rate.beta_ts <- read.csv("yield_rolling_betas_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
115 rownames(rate.beta_ts) <- as.Date(rownames(rate.beta_ts))
116
117 # Time-series of debt-to-asset ratios (dates x tickers)
118 leverage_ts <- read.csv("dtb(leverage)_aligned_final.csv", row.names = 1) %>%
  ↪ as.data.frame()
119 rownames(leverage_ts) <- as.Date(rownames(leverage_ts))
120
121 # Time-series of asset growth / investment rate (dates x tickers)
122 growth_ts <- read.csv("growth_aligned_final.csv", row.names = 1) %>% as.data.frame()
123 rownames(growth_ts) <- as.Date(rownames(growth_ts))
124

```

```

125 # Vanilla market capitalizations
126 # Time-series of Market capitalizations (dates x tickers)
127 mcap <- read.csv("sqrt_mcap_smooth.csv", row.names = 1) %>% as.data.frame()
128 rownames(mcap) <- as.Date(rownames(mcap))
129
130 ts_list <- list(returns = returns,
131               value = value_ts,
132               size = size_ts,
133               momentum = momentum_ts,
134               reversal = reversal_ts,
135               volatility = volatility_ts,
136               illiquidity = illiquidity_ts,
137               yield = yield_ts,
138               oil_beta = oil.beta_ts,
139               mkt_beta = market.beta_ts,
140               rate_beta = rate.beta_ts,
141               leverage = leverage_ts,
142               growth = growth_ts,
143               mcap = mcap) # T x N x K
144
145 ```
146
147 ```{r}
148 # More data cleaning...
149 # We aim for all collected dataframes of the form T x N to have the same structure.
150 # By structure, we seek to have the same collection of NAs and dates across dataframes
151 start_date <- rownames(ts_list[[10]])[1] # First date in shortest object
152 ts_list <- lapply(ts_list, function(x) { # Sets the start date for all other entities
153   x[rownames(x) >= start_date, ]
154 })
155
156
157 # Keep only matching dates across all K time-series
158 all_dates <- Reduce(intersect, lapply(ts_list, rownames))
159 ts_list <- lapply(ts_list, function(x) x[rownames(x) %in% all_dates, ])
160
161 # Keep only common tickers across K characteristic observations
162 all_tickers <- Reduce(intersect, lapply(ts_list, colnames))
163 ts_list <- lapply(ts_list, function(x) x[, colnames(x) %in% all_tickers])
164
165 # Make sure all K time-series have identical NA structure
166 na_union <- is.na(ts_list[[1]])
167
168 for (i in 2:length(ts_list)) {
169   na_union <- na_union | is.na(ts_list[[i]])
170 }
171

```

```

172 ts_list <- lapply(ts_list, function(x) {
173   x[na_union] <- NA
174   x
175 })
176
177 na_first <- is.na(ts_list[[1]])
178
179 ifelse(all(sapply(ts_list, function(x) identical(is.na(x), na_first))),
180        print("All time-series objects share NA structure"),
181        print("Not all time-series objects share NA structure!"))
182
183 # Some rows are still rows filled completely with NAs
184 # All time-series have same NA structure from NA intersect
185 keep_rows <- rowSums(!is.na(ts_list[[1]])) > 0
186
187 cat("dropping", sum(!keep_rows), "NA dates out of", length(keep_rows), "\n")
188
189 ts_list <- lapply(ts_list, function(x) x[keep_rows, , drop = FALSE])
190
191 ```
192
193 ```{r}
194 # Inspect
195 lapply(ts_list, dim)
196 ```
197
198 ```{r}
199 dates <- rownames(ts_list$returns)
200
201 from <- dates[1]
202 to <- dates[length(dates)]
203 sprintf("The final data ranges from %1$s to %2$s", from, to)
204 ```
205
206 # Cross-sectional regression
207
208 ```{r}
209 # Define winsorization function (only keep factor values between the 1st and
210 # -99th percentile for each cross-section).
211 winsorize <- function(x, lower = 0.01, upper = 0.99) {
212   x = as.numeric(x)
213   q = quantile(x, c(lower, upper), na.rm = TRUE)
214   x[x < q[1]] = q[1]
215   x[x > q[2]] = q[2]
216   x
217 }
218

```

```

219
220 # Combine and transform time-series to a collection ordered cross-sections.
221 cs_list <- lapply(1:nrow(ts_list$returns), function(i) {
222
223   # Extract cross-section of mcap, returns and trading volume
224   mcap_i = as.numeric(ts_list$mcap[i,])
225   w = mcap_i / sum(mcap_i, na.rm = TRUE) # Cross-section market cap weights
226
227   returns_i = as.numeric(ts_list$returns[i, ])
228
229   # Leave out mcap, returns and volume
230   factor_id = setdiff(names(ts_list), c("returns", "mcap"))
231
232   # Winsorize -> standardize
233   cs_std <- lapply(factor_id, function(x) {
234
235     factor = as.numeric(ts_list[[x]][i,]) # vector of factor
236     factor = winsorize(factor) # winsorize the factors
237
238     A = factor - sum(w * factor, na.rm = TRUE) # Subtract the market weighted mean
239     B = sqrt(length(factor)) * A / sqrt(sum(A**2, na.rm = TRUE)) # scale
240     B
241   })
242
243   names(cs_std) = factor_id
244   tickers = colnames(ts_list$returns)
245
246   # Store a single cross-section of factors
247   data.frame(
248     returns = returns_i,
249     value = cs_std$value,
250     size = cs_std$size,
251     momentum = cs_std$momentum,
252     reversal = cs_std$reversal,
253     volatility = cs_std$volatility,
254     illiquidity = cs_std$illiquidity,
255     yield = cs_std$yield,
256     oil_beta = cs_std$oil_beta,
257     mkt_beta = cs_std$mkt_beta,
258     rate_beta = cs_std$rate_beta,
259     leverage = cs_std$leverage,
260     growth = cs_std$growth,
261     mcap = mcap_i,
262     row.names = tickers
263   )
264 })
265

```

```

266 cs_dates <- as.Date(rownames(ts_list$returns))
267 names(cs_list) <- cs_dates
268
269 # Cross-sectional regression with mcap weights
270 # The object below contains a collection of linear models, each estimated on a
271 # - cross section for a corresponding date.
272 lm_results_mcap <- lapply(cs_list, function(cs) {
273   lm(
274     returns ~ value + size + momentum + reversal + volatility + illiquidity +
275     yield + oil_beta + mkt_beta + rate_beta + leverage + growth,
276     data = cs,
277     weights = as.numeric(mcap),
278     na.action = na.omit
279   )
280 })
281 ```
282
283
284 ## Residuals / Idioyncratic returns
285
286 ```{r}
287 dates <- as.Date(rownames(ts_list$returns))
288 tickers <- colnames(ts_list$returns)
289
290 T_len <- length(lm_results_mcap)
291 N <- length(tickers)
292
293 res_mat <- matrix(NA_real_, nrow = T_len, ncol = N) # Fill NAs where no residual
294 # is estimated. (example: a ticker were not active at time t)
295 rownames(res_mat) <- dates
296 colnames(res_mat) <- tickers
297
298 for (t in seq_len(T_len)) {
299   reg.fit <- lm_results_mcap[[t]]
300
301   resid <- as.numeric(reg.fit$residuals)
302   used_tickers <- names(reg.fit$residuals)
303
304   col_idx <- match(used_tickers, tickers) # Maintain indexing and naming integrity
305   res_mat[t, col_idx] <- resid
306 }
307
308 res <- as.data.frame(res_mat)
309 rownames(res) <- as.Date(dates)
310 ```
311
312

```

```

313
314
315 # Plot of factor returns
316
317 ## Whole period
318
319
320 ```{r}
321 factor_returns <- t(sapply(lm_results_mcap, coef)) %>% as.data.frame()
322 rownames(factor_returns) <- dates
323 factor_lr <- log(1 + factor_returns) # Convert to log returns
324
325 fr_cum_returns <- apply(factor_lr, 2, cumsum) %>% as.data.frame()
326
327 fr_cum_returns$date <- as.Date(dates)
328
329 # LaTeX does not like underscores "_"
330 fr_cum_returns <- fr_cum_returns %>% rename(
331   `market intercept` = `(Intercept)`,
332   `oil beta` = oil_beta,
333   `market beta` = mkt_beta,
334   `rate beta` = rate_beta
335 )
336
337 # Transform to long format for plotting
338 fr_long <- fr_cum_returns %>% pivot_longer(cols = -date,
339                                           names_to = "factor",
340                                           values_to = "value")
341
342 # Plot factor returns
343 fr_cum_returns_plot_single <-
344   ggplot(fr_long, aes(x = date, y = value, colour = factor)) +
345   geom_line(alpha = 0.8, linewidth = 1) +
346   labs(
347     title = "Daily cumulative factor returns (log returns)",
348     x      = "Year",
349     y      = "Factor returns",
350     colour = "Factors"
351   ) +
352   theme_minimal()
353
354 fr_cum_returns_plot_single
355 ```
356
357 ```{r}
358 tex_plot(fr_cum_returns_plot_single,
359         filename = "single_plot_factor_returns.tex",

```

```

360     width = 6, height = 4)
361 ```
362
363
364 ```{r, fig.width=10, fig.height=7}
365 fr_cum_returns_plot_multi <-
366   ggplot(fr_long, aes(x = date, y = value)) +
367   geom_line(color = "steelblue") +
368   facet_wrap(~ factor, scales = "free_y", ncol = 2) +
369   labs(
370     title = "Daily cumulative factor realisations (log returns)",
371     x      = "Year",
372     y      = "Return"
373   ) +
374   theme_minimal()
375
376 fr_cum_returns_plot_multi
377 ```
378
379
380 ```{r, fig.width=10, fig.height=7}
381 tex_plot(fr_cum_returns_plot_multi, "multi_plot_factor_returns.tex",
382         width = 6, height = 8)
383 ```
384
385 # Average R-Squared
386
387 ```{r}
388 r.sq <- sapply(lm_results_mcap, function(r) {
389   summary(r)$adj.r.squared
390 })
391
392 gen_desc_statistics(r.sq)
393 ```
394
395
396 ```{r}
397 r.sq <- sapply(lm_results_mcap, function(r) summary(r)$adj.r.squared)
398
399 historical_r_squared <- data.frame(
400   date      = as.Date(names(r.sq)),
401   adj_r2    = as.numeric(r.sq)
402 )
403 historical_r_squared$smoothed <-
404   as.numeric(fitted(ses(historical_r_squared$adj_r2, alpha = 1 - 0.95)))
405
406 historical_r.sqr <-

```

```

407 ggplot(historical_r_squared, aes(x = date)) +
408 geom_line(aes(y = adj_r2), alpha = 0.1, linewidth = 0.4) +
409 geom_line(aes(y = smoothed), color = "steelblue", linewidth = 0.9) +
410 labs(title = "Historical R2", x = "Year", y = "R2") +
411 theme_minimal()
412
413 historical_r.sqr
414 ```
415 ```{r}
416 tex_plot(historical_r.sqr, "historical_r_squared_smoothed.tex",
417           width = 6, height = 4)
418 ```
419
420 # Historical Variance Inflation
421
422 ```{r, fig.width=7, fig.height=7}
423 #
424 dates <- rownames(ts_list$returns)
425
426 # Calculate variance inflation per cross-section
427 vif_list <- lapply(lm_results_mcap, VIF) #
428 vif_mat <- do.call(rbind, vif_list)
429 rownames(vif_mat) <- dates
430
431
432 vif_long <- as.data.frame(vif_mat) %>% rename(
433   `oil beta` = oil_beta,
434   `market beta` = mkt_beta,
435   `rate beta` = rate_beta
436 ) %>%
437 mutate(date = as.Date(rownames(vif_mat))) %>%
438 pivot_longer(
439   cols = -date,
440   names_to = "factor",
441   values_to = "vif"
442 )
443
444 # Subplots of Variance inflation of each factor (0-1 very low, 1-5 low,
445 # -5-10 moderate, 10 < severe multicoll)
446 ts_vif <-
447 ggplot(vif_long, aes(x = date, y = vif)) +
448 geom_line(color = "steelblue") +
449 geom_hline(yintercept = 5, linetype = "dashed", color = "red", alpha = 0.7) +
450 facet_wrap(~ factor, scales = "free_y", ncol = 2) +
451 labs(
452   title = "Time-series of daily VIFs per factor",
453   x = "Year",

```

```

454     y      = "Variance Inflation Factor (VIF)"
455   ) +
456   theme_minimal()
457
458 ts_vif
459 ```
460
461
462 ```{r, fig.width=7, fig.height=7}
463 tex_plot(ts_vif, "ts_vif.tex", width = 6, height = 8)
464 ```
465
466
467
468 ```{r}
469 gen_desc_statistics(vif_mat, decimals = 2) %>% t()
470 ```
471
472
473
474 # Hypothesis testing
475
476 ## ADF-test for stationarity
477
478 ```{r}
479 adf_factor <- function(x, k = 0) {
480   x <- as.numeric(x)
481   x <- x[is.finite(x)]
482   test <- adf.test(x, k = k)
483   c(`t-statistic` = unname(test$statistic),
484     `p-value`     = test$p.value)
485 }
486
487
488 adf_results <- t(apply(factor_returns, 2, adf_factor, k = 0)) %>% as.data.frame()
489 adf_results %>% as.data.frame()
490 ```
491
492
493
494 ### Visual inspection
495
496 ```{r}
497 fr <- factor_returns
498 fr$date <- as.Date(dates)
499
500 fr_long <- fr %>% pivot_longer(

```

```

501   cols = -date, names_to = "factor", values_to = "value"
502   )
503
504 ggplot(fr_long, aes(x = date, y = value)) +
505   geom_line(color = "steelblue") +
506   facet_wrap(~ factor, scales = "free_y", ncol = 3) +
507   labs(
508     title = "Daily factor realisations",
509     x      = "Date",
510     y      = "Return"
511   ) +
512   theme_minimal()
513 ```
514
515
516 # Distribution of market capitalizations
517
518 ```{r}
519 nok_mcap <- read.csv("mcap.csv", row.names = 1)
520 rownames(nok_mcap) <- as.Date(rownames(nok_mcap))
521 nok_mcap <- nok_mcap[, colnames(ts_list$returns)]
522
523 mcap_dist_last_date <- data.frame(
524   mcaps = as.numeric(nok_mcap["2025-12-30", ]) / 1e9
525 )
526
527 mcap_dist <-
528   ggplot(mcap_dist_last_date, aes(x = mcaps)) +
529     geom_histogram(bins = 10, fill = "#fd5704", colour = "white") +
530     scale_x_log10(
531       breaks = c(0.1, 1, 10, 100, 1000),
532       labels = c("0.1", "1", "10", "100", "1000")
533     ) +
534     labs(
535       title = "Distribution of market capitalizations (OSEBX, 2025-12-30)",
536       x = "Market capitalization in Billions (NOK) log10 scale",
537       y = "Frequency"
538     ) +
539     theme_minimal()
540
541 mcap_dist
542 ```
543 ```{r}
544 tex_plot(mcap_dist, filename = "mcap_dist.tex", width = 6, height = 4)
545 ```
546
547

```

```

548 # Walk-forward OOS Model comparison of GMV Realized variance
549
550 ```{r}
551 # Extract sufficient sample
552 extract_effective_sample <- function(residuals, # T x N Residuals estimated from
  ↪ iterated WLS
553                                     lambda      # EWMA smoothing parameter
554                                     ) {
555   # This function finds the tickers that were last active and has a sufficient
  ↪ number of observations.
556   # It might seem that this function benefits the characteristic model the
557   # -most since it is based on its residuals.
558   # However, the same return data should exist for the PCA model as well,
559   # -introducing SVD/EM-based imputation to a time-series where missing
560   # -data is usually structural can cause severe bias. The sufficient observation
561   # -limit set through the half-life therefore benefits both models on what
562   # -can be assumed to provide equal footing.
563
564   dates_is = as.Date(rownames(residuals))
565
566   # residuals
567   res_is = residuals %>% as.xts(dates_is)
568
569   half_life = log(0.5) / log(lambda)
570   min_obs = as.integer(half_life * 10) # Rounds float value
571
572   # We must have sufficient observations to calculate the EWMA covariance matrix.
573   # We use the EWMA half-life to create an minimum n-observations rule.
574   # Typical smoothing parameter
575   # Half-life is approximately 13 days.
576   # Tickers sufficiently active near end (must be 10 times half-life
577   # -implied by lambda to be considered sufficient number of obs.)
578
579   last_row = tail(res_is, 1)
580   last_active = colnames(last_row)[!is.na(coredata(last_row))]
581
582   idx_tail = (nrow(res_is) - min_obs + 1):nrow(res_is)
583   n_obs_tail = colSums(!is.na(res_is[idx_tail, last_active]))
584
585
586   good_tickers = names(n_obs_tail)[n_obs_tail >= min_obs]
587
588   good_tickers
589 }
590
591 gmw_weights <- function(factor_cov,          # K x K
592                          idio_var,          # N x N, diagonal entries only

```

```

593         factor_loadings      # N x K
594     ) {
595
596     # This function calculates the GMV portfolio by using the woodbury inverse
597
598     B = factor_loadings
599     ticker_names = rownames(factor_loadings)
600     F_c = factor_cov
601     I = idio_var # must be diagonal
602
603     inv_F = solve(F_c)
604     inv_I = solve(I)
605     X = solve(I, B)
606
607     # Woodbury inverse for numerical stability
608     full_inv.mtx = inv_I - X %*% solve(inv_F + t(B) %*% inv_I %*% B) %*% t(X)
609
610     ones = rep(1, nrow(full_inv.mtx)) # column vector of ones
611
612     over = (full_inv.mtx %*% ones)
613     under = as.numeric(t(ones) %*% full_inv.mtx %*% ones)
614
615     gmv_w = over / under
616     rownames(gmv_w) = ticker_names
617
618     gmv_w
619 }
620
621
622 # Fetch loadings by date
623 loadings_by_date <- function(cs_slice_date,      # YYYY-MM-DD
624                             tickers,      # List of sufficient tickers
625                             factors
626                             ) {
627
628     cs = cs_list[[as.character(cs_slice_date)]]
629     cs$`(Intercept)` = rep(1, nrow(cs))
630
631     # Factor loadings for the cross-section given by (date)
632     B = cs[tickers, factors] %>% as.matrix()
633     B
634 }
635
636
637
638 # Calculate Exponentially smoothed idio variance
639 idio_var_calc <- function(idio_returns,

```

```

640         lambda
641     ) {
642
643     res_list <- lapply(colnames(idio_returns), function(sym) idio_returns[, sym])
644     names(res_list) <- colnames(idio_returns)
645
646     half_life = log(0.5) / log(lambda)
647     min_obs = as.integer(half_life * 10)
648
649     idio_var_ses = sapply(names(res_list), function(ticker) {
650         eps_j = res_list[[ticker]]
651         eps_j = eps_j[!is.na(eps_j)]
652
653         if (length(eps_j) < min_obs) return(NA_real_)
654
655         y = eps_j**2 # Realized idio variance
656
657         # We use the forecast-package Simple exponential smoothing
658         fit = ses(y, h = 1, alpha = (1 - lambda)) # Simple exponential smoothing
659
660         # Exponentially smoothed idiosyncratic variance
661         idio_var = as.numeric(tail(fitted(fit), 1))
662     })
663     # Return smoothed idio variance
664     idio_var_ses
665 }
666
667
668
669 # Perform characteristic risk-decomposition
670 characteristic_gmv <- function(cs_slice_date,      # YYYY-MM-DD
671                               cs_list,            # N x K x T
672                               factor_returns,     # T x K
673                               residual_returns,   # T x K
674                               good_tickers,       # List of tickers with sufficient
675                               ↪ data              # EWMA Smoothing parameter
676                               lambda,             # EWMA Smoothing parameter
677                               mcap
678                               ) {
679
680     dates = as.Date(rownames(factor_returns))
681     sample_idx = dates <= as.Date(cs_slice_date)
682
683     factor_names = names(factor_returns)
684
685     # returns factor cov, loadings, and diag idio variances at time t-1

```

```

686 factor_returns = factor_returns[sample_idx, ] %>% as.matrix()
687
688 idio_returns = residual_returns[sample_idx, good_tickers] %>% as.matrix()
689 stopifnot(identical(colnames(idio_returns), good_tickers))
690
691 # Calculate EWMA factor covariance up to time t-1
692 # Performance analytics EWMA covariance
693 f_cov = M2.ewma(factor_returns, lambda = lambda, as.mat = TRUE)
694
695 # Calculate EWMA factor covariance up to time t-1
696 idio_var = idio_var_calc(idio_returns,
697                           lambda = lambda
698                           ) %>% as.matrix()
699
700 idio_var = diag(as.numeric(idio_var))
701
702 #
703 mcap_vec = as.numeric(mcap[as.character(cs_slice_date), good_tickers])
704 stopifnot(
705   !any(is.na(mcap_vec)),
706   length(mcap_vec) == length(good_tickers)
707 )
708 sqrt_mcap = diag(mcap_vec)
709
710
711 B = loadings_by_date(cs_slice_date, tickers = good_tickers, factor_names)
712 B = B %>% as.matrix()
713 stopifnot(identical(rownames(B), good_tickers))
714
715 # The factor covariance is biased by WLS construction and the use of an incorrect
716   ↪ proxy.
717 # The bias correction can be defined as:
718 bias = solve(t(B) %*% sqrt_mcap %*% B) %*%
719   (t(B) %*% sqrt_mcap %*% idio_var %*% sqrt_mcap %*% B) %*%
720   solve(t(B) %*% sqrt_mcap %*% B)
721
722 unbiased_f_cov = f_cov - bias
723
724 # Guarantee PSD (Eigenvalue clipping)
725 eig <- eigen(unbiased_f_cov, symmetric = TRUE)
726 if (any(eig$values < 0)) {
727   eig$values[eig$values < 1e-10] <- 1e-10 #
728   unbiased_f_cov <- eig$vectors %*% diag(eig$values) %*% t(eig$vectors)
729 }
730
731 gmw_w = gmw_weights(factor_cov = unbiased_f_cov,
732                     idio_var = idio_var,

```

```

732         factor_loadings = B
733     )
734
735     stopifnot(sum(good_tickers != rownames(gmv_w)) == 0)
736
737     gmv_w
738 }
739
740 # Generate exponential weighting matrix for PCA estimation
741 exp_weight <- function(T_len, # Time length
742                        tau      # Exponential smoothing parameter, not to be confused
743                        ↪ with EWMA
744                        ) {
745     t_vec = 1:T_len
746     w = exp(-(T_len - t_vec + 1) / tau)
747     kappa = sqrt(T_len / sum(w**2))
748     kappa * w
749     # returns w where sum(w^2) = T is true. Eigenvalues are scale invariant,
750     # which is why we try to correct the sum of the data back to the original scale
751 }
752
753 # Retrieve model information criterions from PCA factor model
754 factor_IC_criterions <- function(R,
755                                  max_factors = 100
756                                  ) {
757     # Center
758     R = sweep(R, 1, rowMeans(R), "-")
759
760     svd_res = svd(R)
761     U = svd_res$u
762     s = svd_res$d
763     V = svd_res$v
764
765     N = nrow(R)
766     T_len = ncol(R)
767     r_max = min(N, T_len, max_factors, length(s))
768
769     IC1 = numeric(r_max)
770     IC2 = numeric(r_max)
771     IC3 = numeric(r_max)
772
773     for (k in seq_len(r_max)) {
774         U_k = U[, 1:k, drop = FALSE]
775         V_k = V[, 1:k, drop = FALSE]
776         s_k = s[1:k]
777

```

```

778   resid = R - U_k %*% (s_k * t(V_k))
779   NSSR = sum(resid**2) / (N * T_len)
780
781   # Bai & Ng (2002)
782   IC1[k] = log(NSSR) + k * ((N + T_len) / (N * T_len)) * log((N * T_len) / (N +
↪   T_len))
783   IC2[k] = log(NSSR) + k * ((N + T_len) / (N * T_len)) * log(min(N, T_len))
784   # IC2 tended identify the true number of factors in Bai & Ng (2002)
785   IC3[k] = log(NSSR) + k * (log(min(N, T_len)) / min(N, T_len))
786   # IC3 tends to overparameterize
787 }
788
789 data.frame(IC1 = IC1, IC2 = IC2, IC3 = IC3, row.names = seq_len(r_max))
790 }
791
792 # Perform statistical risk-decomposition
793 statistical_gmv <- function(returns,          # T x N
794                             window_size,     # Window
795                             good_tickers,    # List of sufficient tickers
796                             lambda_fast,     # Smoothing parameter for modeling vol
797                             lambda_slow,     # Smoothing parameter for modeling cor
798                             whitening_factors, # Choose n factors for
↪                             heteroskedasticity_remedy
799                             IC               # Choose 1, 2 or 3 (int) for model
↪                             criteria
800                             ) {
801   tot_obs = nrow(returns)
802   window = window_size
803   w_start = tot_obs - window + 1
804   w_end = tot_obs
805   win_idx = w_start:w_end
806
807   # Returns effective window
808   R_win = returns[win_idx, good_tickers]
809
810   # Number of factors for imputation
811   p = min(nrow(R_win), ncol(R_win)) # Max avail factors
812
813
814   pct_missing = (sum(is.na(R_win))) / length(R_win) # Report fraction of missing
↪   values
815
816   # Scaling
817   R_scaled = scale(R_win)
818   R_mu = attr(R_scaled, "scaled:center") # Column means
819   R_sig = attr(R_scaled, "scaled:scale") # Column stds
820

```

```

821 n_imp_factors = round(0.2 * p) # basing 20% giving lowest error in SVDImpute
      ↪ (article)
822
823 R_imp = completeObs(pca(R_scaled, method = "svdImpute", nPcs = n_imp_factors))
824
825 R_imp = sweep(sweep(R_imp, 2, R_sig, "*"), 2, R_mu, "+")
826
827 R = t(R_imp) # Important to note that the returns matrix is now N x T in dimension
828
829 # Data is transposed such that the the singular value decomposition of R returns
830 # - U as the cross sectional dimension, S with singular values, and V with the time
      ↪ dimension.
831
832 # Data dimensions
833 N      = nrow(R)
834 T_len = ncol(R)
835
836
837 # This section establishes weights for estimating idiosyncratic volatility
838 # It is possible to decide how fast volatility spikes occur or
839 # Assume that volatility is fast changing we set the half-life to 3 days
840 # See DCC-GARCH for more information regarding dynamic conditional correlations
841
842 tau_fast = -(1 / log(lambda_fast))
843
844 weights_fast = exp_weight(T_len, tau_fast)
845 weights_fast = diag(weights_fast)
846
847 R_tilde = R %%% weights_fast # N x T
848
849 # First stage SVD
850 # First stage concerns cleaning the data or turning it close to homoscedastic.
851 # -which improve the conditions of factor extraction (Boivin & Ng, 2006).
852 svd_1 = svd(R_tilde)
853
854 # Choose a conservative number of factors for whitening
855 k_factors <- whitening_factors
856
857 U_k = svd_1$u[, 1:k_factors, drop = FALSE] # N x K
858 S_k = svd_1$d[1:k_factors] # K x K, min(N,T) = K
859 V_k = svd_1$v[, 1:k_factors, drop = FALSE] # T x K
860
861 E = R_tilde - U_k %%% (S_k * t(V_k))
862
863 # Estimate idiosyncratic volatility
864 idio_var = rowMeans(E**2)
865 idio_vol = sqrt(idio_var)

```

```

866  idio_vol = pmax(idio_vol, 1e-10) # safe floor idio vol
867  weights_iv = diag(1 / idio_vol)
868
869  # same smoothing as EWMA for the Characteristic model
870  tau_slow = -(1 / log(lambda_slow))
871
872  weights_slow = exp_weight(T_len, tau_slow)
873  weights_slow = diag(weights_slow)
874
875  R_hat = weights_iv %*% R %*% weights_slow
876  R_id = weights_iv %*% R
877
878  # Calculate ICs
879  IC_vals <- factor_IC_criteria(R_id, max_factors = 30)
880
881  stopifnot(IC %in% 1:3)
882
883  p_factors = which.min(IC_vals[[IC]]) # Chosen number of factors after whitening
884
885
886  # Second stage PCA
887  svd_2 = svd(R_hat)
888
889  U_p = svd_2$u[, 1:p_factors, drop = FALSE] # N x p
890  S_p = diag(svd_2$d[1:p_factors]) # p x p
891  V_p = svd_2$v[, 1:p_factors, drop = FALSE] # T x p
892
893  eigenvalues_final <- (svd_2$d**2) / ncol(R_hat)
894
895  noise_bulk = eigenvalues_final[(p_factors+1):length(eigenvalues_final)]
896  mean_noise = mean(noise_bulk)
897
898  eigenv_shrunked = eigenvalues_final
899  eigenv_shrunked[1:p_factors] = pmax(eigenvalues_final[1:p_factors] - mean_noise, 0)
900  eigenv_shrunked[(p_factors+1):length(eigenvalues_final)] = mean_noise
901
902
903  eigen_spikes = eigenv_shrunked[1:p_factors]
904
905  idio_variance = mean_noise * diag(idio_var)
906
907  B = solve(weights_iv) %*% U_p
908  rownames(B) = good_tickers
909
910  gmw_w = gmw_weights(factor_cov = diag(eigen_spikes),
911                      idio_var = idio_variance, factor_loadings = B)
912

```

```

913 factor_var_scaled_back = diag(B %%% diag(eigen_spikes) %%% t(B))
914
915 var_explained = sum(factor_var_scaled_back) /
916   (sum(factor_var_scaled_back) + sum(idio_variance))
917
918 list(weights = gmv_w, k_factors = p_factors,
919       returns_imputed = R_imp, var_explained = var_explained)
920 }
921
922
923 normalize_weights_sum <- function(weights,
924                                   returns
925                                   ) {
926   # It may be that active tickers are delisted at the forecast date.
927   # This function normalizes the forecasted GMV weights_{t-1} so that the
928   # - portfolio is fully invested at time t
929   returns_aligned = returns[rownames(weights)]
930   alive_t = !is.na(returns_aligned) # Listed tickers at forecast horizon t
931   weights_adj = weights[alive_t] / sum(weights[alive_t])
932   sum(weights_adj * returns_aligned[alive_t])
933 }
934
935 sample_gmv = function(returns,
936                       window_size,
937                       good_tickers
938                       ) {
939
940   tot_obs = nrow(returns)
941   win_idx = (tot_obs - window_size + 1):tot_obs
942   R_win = returns[win_idx, good_tickers]
943
944   vanilla_cov = cov(R_win, use = "pairwise.complete.obs")
945
946   # Pairwise covariance may not be postive semi-definite
947   eig = eigen(vanilla_cov, symmetric = TRUE)
948   if (any(eig$values < 1e-10)) {
949     eig$values[eig$values < 1e-10] = 1e-10
950     vanilla_cov = eig$vectors %%% diag(eig$values) %%% t(eig$vectors)
951   }
952
953   vanilla_cov_inv = solve(vanilla_cov + diag(1e-10, nrow(vanilla_cov)))
954   ones = rep(1, nrow(vanilla_cov))
955   w = (vanilla_cov_inv %%% ones) / as.numeric(t(ones) %%% vanilla_cov_inv %%% ones)
956   rownames(w) = good_tickers
957   w
958 }
959

```

```

960 # Walk forward comparison
961 walk_forward_compare <- function(oos_observations = 250,      # Set OOS window
962                                cs_list,                      # N x K x T
963                                returns,                       # T x N
964                                factor_returns,               # T x K
965                                residuals,                     # T x N
966                                pca_window,                    # Window for rolling PCA
967                                lambda,                        # EWMA smoothing parameter
968                                mcap                           # Market cap for
969                                ↪ Characteristic shrinkage
970                                ) {
971 # The goal of the function is to compare OOS GMV-RV of two models (characteristic
972 ↪ and statistical factor model).
973 # The two models decomposes asset risk into systematic and unsystematic risk.
974 # The decomposition be used to estimate the covariance matrix of asset returns, with
975 ↪ more stability than regular sample covariance.
976 # Observations are exponentially weighted, hence one can define an appropriate
977 ↪ half-life
978
979 # Check if dates match
980 dates_r = as.Date(rownames(returns))
981 dates_fr = as.Date(rownames(factor_returns))
982 dates_resid = as.Date(rownames(residuals))
983
984 if (!all(dates_r == dates_fr) || !all(dates_r == dates_resid)) {
985   stop("Dates between returns, factor returns and residuals must be identical!")
986 }
987
988 # Define the effective in-sample and out-of-sample periods
989 return_dates = as.Date(rownames(returns))
990 total_dates = length(return_dates)
991
992 stopifnot(oos_observations < total_dates)
993
994 is_return_dates = return_dates[1:(total_dates - oos_observations)]
995 oos_return_dates = return_dates[(total_dates - oos_observations + 1) : total_dates]
996
997 print_response_is = sprintf("In-sample period: from %1$s to %2$s",
998                             is_return_dates[1],
999                             is_return_dates[length(is_return_dates)])
1000
1001 print_response_oos = sprintf("Out-of-sample period: from %1$s to %2$s",
1002                              oos_return_dates[1],
1003                              oos_return_dates[oos_observations])
1004
1005 is_end_date = is_return_dates[length(is_return_dates)]
1006 oos_start_date = oos_return_dates[1]

```

```

1003 is_end = as.Date(is_end_date)
1004 oos_start = as.Date(oos_start_date)
1005
1006 oos_pos = (length(is_return_dates) + 1) : total_dates
1007
1008 # Empty dataframe for storing realized variances/vol
1009 portfolio_daily_returns = data.frame(
1010     characteristic = rep(NA_real_, oos_observations),
1011     statistical     = rep(NA_real_, oos_observations),
1012     naive           = rep(NA_real_, oos_observations),
1013     char_leverage   = rep(NA_real_, oos_observations),
1014     stat_leverage   = rep(NA_real_, oos_observations),
1015     naive_leverage  = rep(NA_real_, oos_observations),
1016     good_tickers_avail = rep(NA_real_, oos_observations),
1017     stat_N_factors   = rep(NA_real_, oos_observations),
1018     stat_var_explained = rep(NA_real_, oos_observations),
1019     row.names        = as.Date(oos_return_dates)
1020 )
1021
1022 oos_returns = returns[oos_return_dates, ]
1023
1024 # Add progress bar
1025 pb = txtProgressBar(min = 0, max = length(oos_pos), style = 3)
1026
1027 # Extract rolling window
1028 for (i in seq_along(oos_pos)) {
1029     t = oos_pos[i]
1030     is_slice = 1:(t-1)
1031     cs_slice_date = return_dates[t-1]
1032
1033
1034     returns_active = returns[is_slice, , drop = FALSE]
1035     fr_is = factor_returns[is_slice, , drop = FALSE]
1036     res_is = residuals[is_slice,, drop = FALSE]
1037
1038
1039     good_tickers = extract_effective_sample(res_is, lambda)
1040
1041     # Calculate portfolio weights
1042     characteristic_weights = characteristic_gmv(cs_slice_date = cs_slice_date,
1043                                                 cs_list = cs_list,
1044                                                 factor_returns = fr_is,
1045                                                 residual_returns = res_is,
1046                                                 good_tickers = good_tickers,
1047                                                 lambda = lambda,
1048                                                 mcap = mcap
1049     )

```

```

1050
1051 stat_results = statistical_gmv(returns = returns_active,
1052                               window_size = pca_window,
1053                               good_tickers = good_tickers,
1054                               lambda_fast = lambda,
1055                               lambda_slow = lambda,
1056                               whitening_factors = 13,
1057                               IC = 3
1058                               )
1059
1060 stat_weights = stat_results$weights
1061 stat_k_factors = stat_results$k_factors
1062 var_explained = stat_results$var_explained
1063
1064 naive_weights = sample_gmv(stat_results$returns_imputed,
1065                             good_tickers = good_tickers,
1066                             window_size = pca_window
1067                             )
1068
1069 returns_active_t = as.numeric(returns[t, good_tickers])
1070 names(returns_active_t) = good_tickers
1071
1072 # Realized variance at i generated by characteristic portfolio
1073
1074
1075
1076 portfolio_daily_returns$characteristic[[i]] = normalize_weights_sum(
1077     characteristic_weights,
1078     returns_active_t
1079 )
1080
1081 # Realized variance at i generated by statistical portfolio
1082 portfolio_daily_returns$statistical[[i]] = normalize_weights_sum(
1083     stat_weights,
1084     returns_active_t
1085 )
1086
1087
1088
1089 portfolio_daily_returns$naive[[i]] = normalize_weights_sum(
1090     naive_weights,
1091     returns_active_t
1092 )
1093
1094
1095 portfolio_daily_returns$good_tickers_avail[[i]] = length(good_tickers)
1096 portfolio_daily_returns$stat_N_factors[[i]] = stat_k_factors

```

```

1097     portfolio_daily_returns$stat_var_explained[[i]] = var_explained
1098
1099     # Record leverage at time t-1
1100     portfolio_daily_returns$char_leverage[[i]] = sum(abs(characteristic_weights))
1101     portfolio_daily_returns$stat_leverage[[i]] = sum(abs(stat_weights))
1102     portfolio_daily_returns$naive_leverage[[i]] = sum(abs(naive_weights))
1103
1104     cat("Slice date:", cs_slice_date, "\n")
1105     setTxtProgressBar(pb, i)
1106 }
1107
1108 close(pb)
1109
1110 # Return daily GMV-portfolios returns
1111 portfolio_daily_returns
1112 }
1113 ```
1114
1115
1116 ```{r}
1117 nrow(returns[252:nrow(returns), ] )
1118 ```
1119
1120
1121
1122 ```{r, include=FALSE}
1123 returns <- ts_list$returns
1124 rownames(returns) <- as.Date(rownames(returns))
1125
1126 gmv_portfolios_daily <- walk_forward_compare(oos_observations = 1455,
1127                                             cs_list = cs_list,
1128                                             returns = returns,
1129                                             factor_returns = factor_returns,
1130                                             residuals = res,
1131                                             pca_window = 252,
1132                                             lambda = 0.95,
1133                                             mcap = mcap
1134                                             )
1135 ```
1136
1137
1138 # Results
1139
1140 ```{r}
1141 write.csv(gmv_portfolios_daily, file = "gmv_portfolios_daily.csv")
1142 ```
1143

```

```

1144 ```{r}
1145
1146 gmv_portfolios_daily <- read.csv("gmv_portfolios_daily.csv", row.names = 1)
1147
1148 annual_summary <- gmv_portfolios_daily %>%
1149   tibble::rownames_to_column("date") %>%
1150   mutate(
1151     date = as.Date(date),
1152     year = as.integer(format(date, "%Y"))
1153   ) %>%
1154   group_by(year) %>%
1155   summarise(
1156
1157     # Annualized volatility
1158     vol_char          = sd(characteristic, na.rm = TRUE) * sqrt(252),
1159     vol_stat          = sd(statistical, na.rm = TRUE) * sqrt(252),
1160     vol_naive         = sd(naive, na.rm = TRUE) * sqrt(252),
1161
1162     vol_ratio_stat_char = vol_stat / vol_char,
1163     vol_ratio_naive_char = vol_naive / vol_char,
1164
1165     char_leverage = mean(char_leverage),
1166     stat_leverage = mean(stat_leverage),
1167     naive_leverage = mean(naive_leverage),
1168
1169
1170     .groups = "drop"
1171   )
1172
1173 annual_summary
1174 ```
1175
1176 ```{r}
1177 dates <- rownames(gmv_portfolios_daily)
1178
1179 from <- dates[1]
1180 to <- dates[length(dates)]
1181 sprintf("The final data ranges from %1$s to %2$s", from, to)
1182 ```
1183
1184 ```{r}
1185 # Number of tickers that make the effective sample size for each forecast.
1186 sample_n_tickers <- gen_desc_statistics(gmv_portfolios_daily$good_tickers_avail)
1187
1188 sample_n_tickers
1189 ```
1190 ```{r}

```

```

1191 stat_var_explained <- gen_desc_statistics(gmv_portfolios_daily$stat_var_explained)
1192
1193 stat_var_explained
1194 ```
1195
1196 ```{r}
1197
1198 historical_var_explained <- data.frame(
1199   date      = as.Date(rownames(gmv_portfolios_daily)),
1200   var_exp   = as.numeric(gmv_portfolios_daily$stat_var_explained)
1201 )
1202 historical_var_explained$smoothed <- as.numeric(fitted(ses(
1203   gmv_portfolios_daily$stat_var_explained, alpha = 1 - 0.95
1204 )))
1205
1206 historical_var_explained_plot <-
1207   ggplot(historical_var_explained, aes(x = date)) +
1208   geom_line(aes(y = var_exp), alpha = 0.1, linewidth = 0.4) +
1209   geom_line(aes(y = smoothed), color = "steelblue", linewidth = 0.9) +
1210   labs(
1211     title = "Historical Variance Explained",
1212     x = "Year",
1213     y = "Variance Explained") +
1214   theme_minimal()
1215
1216 historical_var_explained_plot
1217 ```
1218
1219 ```{r}
1220 tex_plot(historical_var_explained_plot,
1221   filename = "hist_var_explained.tex", width = 6, height = 4)
1222 ```
1223
1224
1225 ```{r}
1226 annual_summary <- gmv_portfolios_daily %>%
1227   tibble::rownames_to_column("date") %>%
1228   mutate(
1229     date = as.Date(date),
1230     year = as.integer(format(date, "%Y"))
1231   ) %>%
1232   group_by(year) %>%
1233   summarise(
1234     vol_char = sd(characteristic, na.rm = TRUE) * sqrt(252),
1235     vol_stat = sd(statistical,      na.rm = TRUE) * sqrt(252),
1236     vol_naive = sd(naive,          na.rm = TRUE) * sqrt(252),
1237     vol_ratio_stat_char = vol_stat / vol_char,

```

```

1238     vol_ratio_naive_char = vol_naive / vol_char,
1239     lev_char   = mean(char_leverage, na.rm = TRUE),
1240     lev_stat   = mean(stat_leverage, na.rm = TRUE),
1241     lev_naive  = mean(naive_leverage, na.rm = TRUE),
1242     .groups = "drop"
1243 )
1244
1245 annual_summary
1246 ```
1247
1248 ```{r}
1249 n_factor_est_data <- rowSums(!is.na(res))
1250 n_factor_est_data <- gen_desc_statistics(n_factor_est_data)
1251
1252 # The forecast sample
1253 n_forecast_data <- gen_desc_statistics(gmv_portfolios_daily$good_tickers_avail)
1254
1255 # difference between factor return estimation and
1256 # -effective sample of naive and statistical model
1257 sample_diff <- n_factor_est_data - n_forecast_data
1258
1259 ```
1260
1261
1262 ```{r}
1263
1264 df <- data.frame(
1265   date      = as.Date(rownames(gmv_portfolios_daily)),
1266   loss_diff = (gmv_portfolios_daily$characteristic**2) -
1267     ↪ (gmv_portfolios_daily$naive**2)
1268 )
1269
1270 lambda <- 0.95
1271 fit <- ses(df$loss_diff, alpha = 1 - lambda, h = 1)
1272
1273 df$ses_smooth <- as.numeric(fitted(fit))
1274
1275 c_n <-
1276   ggplot(df, aes(x = date)) +
1277   geom_line(aes(y = ses_smooth), color = "black", linewidth = 0.4) +
1278   geom_hline(yintercept = 0, linetype = "dashed") +
1279   labs(
1280     title = sprintf("Loss Difference  (%1$s - %2$s)", "Char2", "Naive2"),
1281     y = "Difference",
1282     x = "Year"
1283   )

```

```

1284 c_n
1285 ```
1286
1287 ```{r}
1288 tex_plot(c_n, filename = "charVSnaive.tex", width = 6, height = 4)
1289 ```
1290
1291
1292 ```{r}
1293 df <- data.frame(
1294   date      = as.Date(rownames(gmv_portfolios_daily)),
1295   loss_diff = (gmV_portfolios_daily$statistical**2) - (gmV_portfolios_daily$naive**2)
1296 )
1297
1298 lambda <- 0.95
1299 fit    <- ses(df$loss_diff, alpha = 1 - lambda, h = 1)
1300
1301 df$ses_smooth <- as.numeric(fitted(fit))
1302
1303 s_n <-
1304   ggplot(df, aes(x = date)) +
1305   geom_line(aes(y = ses_smooth), color = "black", linewidth = 0.4) +
1306   geom_hline(yintercept = 0, linetype = "dashed") +
1307   labs(
1308     title = sprintf("Smoothed Loss Difference  (%1$s - %2$s, lambda = %3$s)", "Stat2",
1309       ↪ "Naive2", lambda),
1310     y = "Difference",
1311     x = "Year"
1312   )
1313
1314 s_n
1315 ```
1316 tex_plot(s_n, filename = "statVSnaive.tex", width = 6, height = 4)
1317 ```
1318
1319
1320 ```{r}
1321 df <- data.frame(
1322   date      = as.Date(rownames(gmv_portfolios_daily)),
1323   loss_diff = (gmV_portfolios_daily$characteristic**2) -
1324     ↪ (gmV_portfolios_daily$statistical**2)
1325 )
1326
1327 lambda <- 0.95
1328 fit    <- ses(df$loss_diff, alpha = (1 - lambda), h = 1)
1329

```

```

1329 df$ses_smooth <- as.numeric(fitted(fit))
1330
1331 c_s <-
1332   ggplot(df, aes(x = date)) +
1333     geom_line(aes(y = ses_smooth), color = "black", linewidth = 0.4) +
1334     geom_hline(yintercept = 0, linetype = "dashed") +
1335     labs(
1336       title = sprintf("Smoothed Loss Difference  (%1$s - %2$s, lambda = %3$s)", "Char2",
1337         ↪ "Stat2",lambda),
1338       y = "Difference",
1339       x = "Year"
1340     )
1341 c_s
1342 ```
1343 ```{r}
1344 tex_plot(c_s, filename = "charVSstat.tex", width = 6, height = 4)
1345 ```
1346
1347
1348
1349 ```{r}
1350 lambda <- 0.95
1351 smoothed_RV_char <- ses(gmv_portfolios_daily$characteristic**2,
1352   alpha = 1 - lambda, h = 1) %>% fitted() %>% as.numeric()
1353
1354 smoothed_RV_stat <- ses(gmv_portfolios_daily$statistical**2,
1355   alpha = 1 - lambda, h = 1) %>% fitted() %>% as.numeric()
1356
1357 smoothed_RV_naive <- ses(gmv_portfolios_daily$naive**2,
1358   alpha = 1 - lambda, h = 1) %>% fitted() %>% as.numeric()
1359
1360 real_var <- data.frame(
1361   date      = as.Date(rownames(gmv_portfolios_daily)),
1362   char = smoothed_RV_char,
1363   stat = smoothed_RV_stat,
1364   naive = smoothed_RV_naive
1365 )
1366
1367
1368 rv_long <- real_var %>% pivot_longer(cols = -date, names_to = "model", values_to =
1369   ↪ "value")
1370
1371 cns <-
1372   ggplot(rv_long, aes(x = date, y = value, colour = model)) +
1373     geom_line(alpha = 0.8, linewidth = 0.5) +
1374     labs(

```

```

1374     title = "Smoothed Daily Realized Variance (lambda = 0.95)",
1375     x      = "Year",
1376     y      = "RV",
1377     colour = "Model"
1378   ) +
1379   theme_minimal()
1380
1381   cns
1382   ```
1383   ```{r}
1384   tex_plot(cns, filename = "smoothed_rv.tex", width = 6, height = 4)
1385   ```
1386   ```{r}
1387
1388   real_var <- data.frame(
1389     date      = as.Date(rownames(gmv_portfolios_daily)),
1390     char = gmv_portfolios_daily$characteristic**2,
1391     stat = gmv_portfolios_daily$statistical**2,
1392     naive = gmv_portfolios_daily$naive**2
1393   )
1394
1395
1396   rv_long <- real_var %>% pivot_longer(cols = -date,
1397                                         names_to = "model", values_to = "value")
1398
1399   cns_ins <-
1400     ggplot(rv_long, aes(x = date, y = value, colour = model)) +
1401     geom_line(alpha = 0.8, linewidth = 0.5) +
1402     labs(
1403       title = "Daily Realized Variance",
1404       x      = "Year",
1405       y      = "Relized Variance",
1406       colour = "Model"
1407     ) +
1408     theme_minimal()
1409
1410   cns_ins
1411   ```
1412   ```{r}
1413   tex_plot(cns_ins, filename = "instant_rv.tex", width = 6, height = 4)
1414   ```
1415
1416
1417   ```{r}
1418   df <- data.frame(
1419     date      = as.Date(rownames(gmv_portfolios_daily)),
1420     factors_chosen = gmv_portfolios_daily$stat_N_factors

```

```

1421 )
1422
1423 stat_n_fact <-
1424   ggplot(df, aes(date, factors_chosen)) +
1425   geom_line() +
1426   labs(
1427     title = "Statistical Model: Daily Number of Factors Chosen",
1428     y = "Factors",
1429     x = "Year"
1430   )
1431
1432 stat_n_fact
1433 ```
1434
1435 ```{r}
1436 tex_plot(stat_n_fact, filename = "stat_n_fact.tex", width = 6, height = 4)
1437 ```
1438
1439
1440 ```{r}
1441 dm.test(gmv_portfolios_daily$characteristic, gmv_portfolios_daily$naive, power = 2)
1442 dm.test(gmv_portfolios_daily$statistical, gmv_portfolios_daily$naive, power = 2)
1443 dm.test(gmv_portfolios_daily$characteristic, gmv_portfolios_daily$statistical, power =
1444   ↪ 2)
1445 ```
1446
1447 ```{r}
1448 dm_CN <- dm.test(gmv_portfolios_daily$characteristic,
1449                 gmv_portfolios_daily$naive,          power = 2)
1450
1451 dm_SN <- dm.test(gmv_portfolios_daily$statistical,
1452                 gmv_portfolios_daily$naive,          power = 2)
1453
1454 dm_CS <- dm.test(gmv_portfolios_daily$characteristic,
1455                 gmv_portfolios_daily$statistical, power = 2)
1456
1457 dm_table <- data.frame(
1458   Models = c("Characteristic vs. Naive",
1459              "Statistical vs. Naive",
1460              "Characteristic vs. Statistical"),
1461   `DM statistic` = c(dm_CN$statistic, dm_SN$statistic, dm_CS$statistic),
1462   `p-value`      = c(dm_CN$p.value,   dm_SN$p.value,   dm_CS$p.value)
1463 )
1464
1465 dm_table
1466 ```

```

```

1467 ```{r}
1468 # The original OOS-test, with saved results from the first version
1469 gmv_portfolios_daily <- read.csv("gmv_portfolios_daily_v1.csv", row.names = 1)
1470
1471 dm_CS <- dm.test(gmv_portfolios_daily$characteristic,
1472                 gmv_portfolios_daily$statistical, power = 2)
1473
1474 dm_table <- data.frame(
1475   Models = c( "Characteristic vs. Statistical"),
1476   `DM statistic` = c(dm_CS$statistic),
1477   `p-value`      = c(dm_CS$p.value)
1478 )
1479
1480 dm_table
1481 ```
1482
1483 ```{r}
1484 annual_vol_char = sd(gmv_portfolios_daily$characteristic) * sqrt(252)
1485 annual_vol_stat = sd(gmv_portfolios_daily$statistical) * sqrt(252)
1486 annual_vol_naive = sd(gmv_portfolios_daily$naive) * sqrt(252)
1487
1488 cat("Characteristic Annual Vol 2025:", annual_vol_char, "\n")
1489 cat("Statistical Annual Vol 2025:", annual_vol_stat, "\n")
1490 cat("Naive Annual Vol 2025:", annual_vol_naive, "\n")
1491 ```
1492 ```{r}
1493 gmv_portfolios_daily <- read.csv("gmv_portfolios_daily.csv", row.names = 1)
1494
1495 loss_diff <- data.frame(
1496   row.names = rownames(gmv_portfolios_daily),
1497   `CN (Diff)` = gmv_portfolios_daily$characteristic**2 -
1498     ↪ gmv_portfolios_daily$naive**2,
1499   `SN (Diff)` = gmv_portfolios_daily$statistical**2 - gmv_portfolios_daily$naive**2,
1500   `CS (DIFF)` = gmv_portfolios_daily$characteristic**2 -
1501     ↪ gmv_portfolios_daily$statistical**2
1502 )
1503
1504 # Check if loss-differentials are at last stationary
1505 adf_factor <- function(x, k = 0) {
1506   x <- as.numeric(x)
1507   x <- x[is.finite(x)]
1508   test <- adf.test(x, k = k)
1509   c(`t-statistic` = unname(test$statistic),
1510     `p-value`     = test$p.value)
1511 }

```

```
1512 adf_results <- t(apply(loss_diff, 2, adf_factor, k = 0)) %>% as.data.frame()
1513
1514 ```
1515
```